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24 May 1982
SUPERSEDING
MIL-T-52930A
24 June 1980

MILITARY SPECIFICATION

TRUCK, LIFT, FORK, ELECTRIC, FRONT/SIDE LOADING

SIT-DOWN, 4000/3000-POUND CAPACITY

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a truck having a front load capacity of 4000 pounds at 24 inch load center and a side load capacity of 3000 pounds at 24 inch load center. The truck shall have front wheel drive, rear wheel steer, telescopic mast, power steering, hydraulic brakes, solid rubber tires and overhead guard. The truck shall have automotive type steering.

1.2 Classification. Trucks shall be of the following type construction, in accordance with ANSI/UL583, and styles, as specified (see 6.2):

Type E - Nonspark enclosed.
  Style 1 - Collapsed mast height/lift height: 89 inches maximum/198 inches minimum.

Type EE - Spark enclosed
  Style 2 - Collapsed mast height/lift height: 83 inches maximum/180 inches minimum.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Mobility Equipment Research and Development Command, ATTN: DRDME-DS, Fort Belvoir, Virginia 22060 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 3930

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2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified (see 6.2), the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-V-173	- Varnish, Moisture- and Fungus-Resistant (For Treatment of Communications, Electronic, and Associated Equipment).
MIL-P-514	- Plates, Identification, Instruction, and Marking, Blank.
MIL-G-3859	- Grease Guns, Hand-Operated, Lever, Push and Screw Type.
MIL-M-3971	- Meter, Time Totalizing, Non-Hermetically Sealed, Electrical: General Specification for.
MIL-B-46176	- Brake Fluid, Silicone, Automotive, All Weather Operational and Preservative.

STANDARDS

FEDERAL

FED-STD-595	- Colors.

MILITARY

MIL-STD-130	- Identification Marking of US Military Property.
MIL-STD-162	- Materials Handling Equipment: Preparation for Shipment, Storage, Cyclic Maintenance, Routine Testing and Processing.
MIL-STD-209	- Slinging and Tiedown Provisions For Lifting and Tying Down Military Equipment.
MS51336	- Lunette - Coupler, Drawbar, Rings.

(Copies of specifications and standards required by manufacturers in connection with specific acquisition functions should be obtained from the procuring activity or as directed by the contracting officer).

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2.1.2 Other Government documents, drawings, and publications. The following other Government documents form a part of this specification to the extent specified herein.

DEPARTMENT OF LABOR, OCCUPATIONAL SAFETY AND HEALTH ADMINISTRATION

29 CFR, Chapter XVII, Part 1910.
Occupational Safety and Health Standards

(Application for copies should be addressed to the Superintendent of Documents, U. S. Government Printing Office, Washington, DC 20402.)

2.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

Safety Standard For Powered Industrial Trucks (ANSI B56.1)
Standard for Electric-Battery-Powered Industrial Trucks (ANSI/UL583)
Load Handling Symbols for Powered Industrial Trucks (MH 11.3)
Forks and Fork Carriers for Powered Industrial Trucks (MH 11.4)

(Application for copies should be addressed to the American National Standards Institute, 1430 Broadway, New York, NY 10018.)

NATIONAL ELECTRICAL MANUFACTURERS ASSOCIATION (NEMA)

NEMA Standards

(Application for copies should be addressed to the National Electrical Manufacturers Association, 155 East 44th Street, New York, NY 10017.)

SOCIETY OF AUTOMOTIVE ENGINEERS (SAE)

SAE Handbook

(Application for copies should be addressed to the Society of Automotive Engineers, 400 Commonwealth Drive, Warrendale, PA 15096.)

TIRE AND RIM ASSOCIATION, INC. (TRA)

TRA Yearbook

(Application for copies should be addressed to the Tire and Rim Association, Inc., 3200 West Market Street, Arkron, OH 44313.)

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SOUTHERN CALIFORNIA AIR POLLUTION CONTROL DISTRICT

Rule 102 - Photochemically Reactive Solvents.

(Application for copies should be addressed to the Southern California Air Pollution Control District, 434 South San Pedro Street, Los Angeles, CA 90013.)

(Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Description. The truck shall be electric motor driven, front wheel drive, rear wheel steer, sit-down, front/side loading type. Truck capacity shall be 4000 pounds at a 24 inch load center to full lift height when operating as a counterbalanced front loading unit. As a counterbalanced side loading unit, the capacity shall be 4,000 pounds at a 24 inch load center to a lift height of 72 inches minimum and down rated to 3,000 pounds at a 24 inch load center to full lift height. Truck requirements shall be as specified herein.

3.2 First article (first-produced truck). The contractor shall furnish one or more trucks as specified (see 6.2), for examination and testing within the time frame specified (see 6.2) to prove that trucks to be furnished comply to the requirements of this specification. Examination and tests shall be as specified in Section 4 and shall be subject to surveillance and approval by the Government.

3.3 Safety. The truck shall conform to the applicable requirements of ANSI B56.1 and UL583 for type E trucks and type EE trucks, and such additional requirements as specified herein, with the Occupational Safety and Health Standards requirements of 29 CFR, Chapter XVII, Part 1910 and all pertinent national safety standards in effect at the time of solicitation.

3.4 Human factors engineering. The characteristics of the truck shall provide for operation by personnel ranging from the small person, clothed, through the large person, clothed, in accordance with SAE J833.

3.5 Design. Design of trucks covered by the specification and all components therein shall insure safe operation when operated as specified herein. There shall be no evidence of accelerated wear, failures, or permanent deformation when operated as specified herein. There shall be no exposed bolts, clamps, gages, fittings, lifting parts, or appendages that can be caught or hooked while the truck is working in a confined space.

3.5.1 Materials. Material shall be as specified herein. Materials not specified shall be selected by the contractor and shall be subject to all provisions of the specification (see 6.5).

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3.5.2 Lubrication fittings. Lubrication fittings shall conform to SAE J534. Fittings shall be located in a protected position and shall be accessible to a hand-operated grease gun conforming to MIL-G-3859 with a 10-inch flexible extension. Accessibility to fittings shall be provided without the removal or adjustment of accessories or parts. Panels and plates equipped with hand-operable, quick-disconnect fasteners or not more than eight common fasteners may be used to provide accessibility.

3.6 Maintainability. Provision shall be made for adjustment, servicing, and replacement of all electrical assemblies and components, hydraulic system components, battery, wearing parts of lift and tilt mechanism brakes and components, wear parts of steering assembly, tires, wheels, lights, and horn. If hand access openings are required to perform maintenance, the edges of each opening shall be smooth and the opening shall be provided with a removable or hinged cover. Dimensions of access openings shall be in accordance with SAE J925 for bare hand. There shall be no interference with the servicing or drawing of oils and lubricants to or from any assembly or components by frame members or other obstructions, and without requiring the use of flexible connections all resevoirs with removable drain plugs shall provide for complete drainage. Unless otherwise specified (see 6.2), all special handtools required for maintenance of the truck shall be provided and shipped with each truck procured. The truck shall be designed so as not to require preventive maintenance or repair services at intervals shorter than 2000 laps when tested as specified in 4.5.29 except for servicing truck batteries and removing foreign matter from tire treads. Reservoir level checks are not considered scheduled maintenance. The ratio of man hours of scheduled and unscheduled maintenance required to the hours of operation performed shall not exceed .08.

3.7 Maintenance operations. Each operation in the following maintenance operations list shall be accomplished by one man in not more than the time specified, using common tools and any special tools, furnished with the truck:

Maintenance Operation	(Time in Minutes)
(a) Remove and replace drive motor brushes	90
(b) Remove and replace hoist and tilt motor brushes	90
(c) Remove and replace steer motor brushes	90
(d) Remove and replace all contactor tips	90
(e) Remove and replace all filters, screens and strainers in hydraulic system	105
(f) Remove and replace shoe lining, spring, cylinder and any other brake component. Bleed and refill master cylinder (two men)	45

3.8 Electrical system.

3.8.1 Enclosure and shields. All electrical components including wiring shall be so located or protected by adequate means to prevent contact by excess moisture or battery electrolyte in the event of spillage.

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3.8.2 Overcurrent devices. In addition to the requirements of 3.3, overcurrent devices shall be provided on each power and each control circuit.

3.8.3 Static discharge device. Trucks shall be equipped with two replaceable electrically conductive rubber straps, mounted immediately adjacent to the drive wheels and the resistance shall be in conformance to ANSI/UL583.

3.8.4 Instrument panel. The instrument panel shall be designed for easy removal as a unit or front removal of electrical assemblies mounted thereon. Mounting studs, bolts, or nuts, shall be so secured that they will not turn or become loose when an assembly is removed from the panel. The instrument panel cover shall be unobstructed to permit removal of the cover for inspection and servicing without removal of any major part or interference from truck structure. The instrument panel shall be designed to provide an unobstructed view of all gages on the panel.

3.8.5 Wiring. All wiring shall be color or number coded and the indentification scheme shall be included on the electric diagram.

3.9 Propulsion and hydraulic systems.

3.9.1 Motor and motor accessories.

3.9.1.1 Motors. Traction motors shall be industrial type, with a high starting torque, designed for efficient operation on 36 volts, direct current. Motors shall be capable of withstanding the stresses and current loads induced by tests required by this specification. The resultant strains and temperature rise shall not cause failure of any component part or excessive pitting other than normal discoloration of the commutators. The hydraulic pump and steering pump motor, if used, shall successfully withstand the stresses and current loads induced by tests required by this specification. Motor(s) for type EE trucks shall be provided with one or more thermal relay switches which shall cause disconnection of battery power to the affected circuit before the outside surface temperature of the motor(s) exceeds 347° F. Thermal relays shall be located in critical heat areas and shall interupt the current only to the overheated circuit.

3.9.1.2 Insulation. Insulation for motors shall be Class B, F or H as defined in NEMA Standards except the insulation shall not contain silicone materials. Silicone material in any form shall not be used in the motor. The electrical and mechanical properties of the insulated windings shall not be impaired by the application of the temperature permitted for the class insulation used (the word "impaired" is here used in the same sense of causing any change which could disqualify the insulating material for continuous service).

3.9.1.3 Speed controller. The speed control shall be solid state type. The controls shall be of heavy duty design for continuous truck use and shall not incur failure or damage when the truck is tested in accordance with Section 4. The controller shall provide means to permit precision maneuvering in restricted areas. It shall provide stepless or combination step and stepless acceleration through the complete speed range from zero to full speed in both the forward and

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reverse directions and shall have automatic electrical or mechanical interlocks to prevent starting except in low speed. It shall provide automatic means of dynamic braking to limit the plugging when the controller is reversed with the truck in motion. All contactors on a given truck shall be designed to provide maximum interchangeability of parts. Contactors shall be designed for operation on 36 volts direct current and shall have easily replaceable tips of copper or silver alloy composition. Replacement of contactor tips shall be accomplished by use of handtools only. There shall be no evidence of failure in any part of the controller when the truck undergoes tests specified herein.

3.9.1.4 Safety cutout system.

3.9.1.4.1 Type E. Provisions shall be made for type E trucks that all power to the traction motor is interrupted when the operator leaves his seat.

3.9.1.4.2 Type EE. In order to insure that no current is flowing when the connector is being disconnected on Type EE trucks, a magnetic contactor shall be installed in one of the two main power lines from the battery connector. The control circuit for the magnetic contactor shall be wired with two switches in series, either of which shall cause the contactor to open and disconnect all power. The first switch shall be a seat switch. The second switch shall be an emergency switch with a positively attached mushroom type button located within easy reach of the operator and painted red.

3.9.2 Battery. The trucks shall be furnished with industrial lead acid battery having sufficient capacity to enable the truck to meet the requirements of 4.5. Battery leads shall be equiped with a two contact, type EC locking half connector, that is used to connect with the electric motor through the mounting half connector on the truck or to connect to the battery charger through the mounting half connector on the charger. When first-produced truck is specified (see 6.2), the battery for the first-produced truck used for testing by the contractor, shall be shipped in a wet and charged condition after the exterior of the battery has been rehabilitated to a "like new" appearance.

3.9.2.1 Battery connector mounting. Trucks shall be equipped with standard EC mounting half connectors, suitably located to fit the locking half as specified in 3.9.2.

3.9.3 Battery compartment. Battery compartment dimensions shall have a 1/4-inch-minimum clearance maintained on all sides and a minimum of 1/2-inch clearance at the top of the battery. The compartment shall be constructed to permit quick inspection and servicing of battery using simple handtools as well as easy battery removal. Effective means shall be provided to secure battery in place. The securing device shall be attached so that it can be easily removed with the use of handtools. The battery compartment shall permit removal or installation of the battery from one side or top of the truck. The underside of the compartment cover shall be lined with a sheet of 1/4-inch plywood finished with not less than three coats of asphalt paint, or equivalent insulating material. The inside of the battery compartment shall be finished with an acid resistant paint to resist deterioration through normal contact with electrolyte. The outside of the entire compartment shall be free of all markings.

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3.9.4 Drive unit. The drive shall be a gear reduction unit(s) connected to the driving motor(s), totally enclosed and operating in a constant oil bath. The final gear reduction may be in a totally enclosed casing which shall provide positive periodic lubrication of gear teeth. Drive gear housing(s) shall be equipped with filler and magnetic drain plug. Provision shall be made for adjustment where necessary to compensate for wear in the differential and pinion. All required drain plugs, check plugs, or oil level checking and filling devices shall be readily accessible without removing major components.

3.9.5 Hydraulic system. The hydraulic system shall consist of all hydraulic components necessary for operation of the forks and mast and does not include brake and steering system components. The system shall have a maximum working pressure (relief valve setting) not in excess of 2500 psi. The hydraulic system components shall be capable of withstanding 300 percent rated load without leakage or damage. All system components which are subject to full system pressure shall withstand a proof pressure that is equal to or greater than 1-1/2 times the maximum working pressure (1-1/2 times relief value setting) without external leakage, damage, or permanent deformation. Hydraulic fittings on pumps, valves, and cylinder shall be the O-ring straight thread type as specified in SAE J514 or 4-bolt flanged type as specified in SAE J518. The system shall provide for lowering of rated load at the rate specified in 3.24.4. A pipe tap or straight thread O-ring boss test point shall be located at the outlet of each pump or pressure inlet of each control valve for measuring pressure. A system filter shall be furnished in either the suction or return line. Hose reels shall be mounted such that they are within the plan outline of the truck.

3.9.5.1 Hydraulic lines. Hydraulic lines shall consist of tubing, hose and fittings. Pressure and suction hose shall conform to SAE J517. Field attachable reuseable fittings shall be compatible with the truck design such that they will replace each permently attached coupling, and the replacement hose assembly shall fit in place of the orginal hose. Pressure fittings shall be either 37-degree flare, female swivel or 4-bolt split flange as specified in SAE J516. The hose selected shall have a working pressure which is equal to or greater than the hydraulic system maximum relief value setting. Pressure tubing shall be in accordance with SAE J524 or SAE J525. Pressure tube fittings shall be 37-degree flare on "O" ring boss conforming to SAE J518. Pressure tubing and tube fittings shall have a minimum burst pressure that is equal to or greater than 3.5 times the maximum system working pressure.

3.9.5.2 Reservoir. A reservoir shall be furnished to contain the hydraulic system oil. The reservoir shall include a filler opening with cap. The location shall not impair operational visibility or access to either side of the operator's compartment. The reservoir shall have sufficient capacity to prevent air entering the system with all hydraulic pistons fully extended and sufficient free air capacity to prevent oil being discharged through the reservoir air vent when maximum return flow of oil is surged into the reservoir from the system. A dip stick shall be provided and marked to indicate full and add positions. The filler tube shall be not less than 1 inch inside diameter. A breather with a maximum rating of 15 micron shall be provided and shall filter incoming air. The breather shall preclude the entrance of water induced by rain. The reservoir filler neck shall contain a removable 30 mesh strainer retained in place unless all oil added

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to the reservoir is filtered by an equal or finer filter prior to passing through the pump. The reservoir shall have one or more accesses for manual cleaning of the entire inside or be removable for cleaning. A removable strainer, 50 mesh minimum, shall be installed in the suction line between the reservoir and the pump unless a suction line filter is furnished. Two or more thermal switches shall be installed in different critical heat areas on the hydraulic reservoir and each one, separately, shall cause disconnection of the pump motor(s) before the outside surface temperature of the hydraulic reservoir exceeds 347° F.

3.9.5.3 Cylinders. Cylinders shall be so located as to provide for ease of maintenance and replacement. All internal cylinder ports shall be located beyond the area of piston travel. Each cylinder shall have an exclusion device to prevent foreign material and fluids from entering around the piston rod and damaging the seal and other cylinder components. The rod shall be hard chrome plated to prevent corrosion. After plating, maximum rod surface roughness shall not exceed 32 microinches. All hydraulic system and steering cylinders shall withstand a pressure that is equal to 1.5 times the system relief valve setting without evidence of rupture, permanent deformation, damage, or external leakage. The bearing surface shall be of a material that will not scar, score, or gall the inside surface of the cylinder or the cylinder rod.

3.9.5.4 Contamination levels for hydraulic system. When the hydraulic system is tested as specified in 4.5.26 particle counts shall not exceed 1000 particles per milliliter greater than 10 micrometers, and particle counts shall not exceed 10 particles per milliliter greater than 20 micrometers.

3.10 Structure. The frame, fork, fork carriage, load backrest and uprights shall be capable of withstanding 300 percent rated load without permanent deformation.

3.10.1 Frame. Frame members, bracings, and all their joints shall provide a rigid unit structure. The frame and related structure shall have sufficient depth to afford protection for working parts and shall be capable of withstanding, without permanent deformation, stresses induced by and tests in Section 4.

3.10.2 Width. The overall width of the vehicle, at the widest point, shall not exceed 46 inches for style 1 truck and 44 inches for style 2 truck.

3.10.3 Height. The overall truck height, with mast retracted, shall not exceed 89 inches for style 1 truck and 83 inches for style 2 truck.

3.10.4 Uprights and carriage assembly. Uprights shall be of the telescopic, roller type. Rollers shall be of the permanently lubricated for life type. Uprights shall be positioned such that the rear of the vertical upright of the fork carriage are forward of the plane of the front face of the drive tires. The lift cylinder shall be mounted and positioned so that it does not protrude forward of the plan outline of the uprights. Chains shall be installed in such a manner that they cannot be displaced from the roller until they have been raised a minimum of 3/8 inch. The lift shall be hydraulically actuated, with the upper portion automatically lifting as the load is raised above the "free lift" position. Means shall be provided in the uprights to compensate by replacement or adjustment for

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wear which may develop between lateral rolling contacts. Positive means shall be provided to prevent over-travel of the carriage or channel in both upper and lower positions. Uprights shall be capable of withstanding the tests specified in 4.5.2

3.10.4.1 Forks and fork carriage. The forks and fork carrier shall conform to ANSI B56.1 and MH 11.4. A cutout shall be provided in the fork carriage so that the lower hook of the fork will fit through the slot and each fork shall be removable without removing the load backrest. Forks shall be 48 inches ± 1/8 inch long for style 1 truck and shall be not less than 38 inches nor more than 40 inche long for style 2 truck. Width of forks shall not be less than 4 inches, nor more than 5-1/2 inches. Fork thickness shall not exceed 1/2 inch at the tips and the end taper shall be not less than 14 inches long. Taper shall be on the underside of the forks. The maximum fork spread shall be 37-inches measured between outer edges of the forks. The top of the fork tips shall be chamfered or rounded to break corners and provide smooth edges. Forks on trucks shall be adjustable laterally without the use of handtools. The pivot, transverse, lift, and lower movements shall be capable of independent operation and shall also be capable of simultaneous operation of any to all combinations. The capability shall be available with the truck at stand still and in motion. The operator shall be able to see the tip of at least one fork at any lift height in either the forward or side leading position without leaving his seat when there is no load on the forks. Forks for trucks shall show no evidence of cracks or failure when inspected in accordance with Test Method No. 9.

3.10.4.2 Load backrest. A metal load backrest, removable without disturbing an other component, shall be provided. There shall be no protruding bolts or appendages beyond the side plane of the load backrest. The load backrest in conjunction with fork and hanger design shall provide a vertical rear guard at least 48 inches high measured from load carrying surfaces of the forks. It shall have width and capacity and strength sufficient to prevent a rated load 48 inches high or less from forcing backrest against the uprights when the mast is at maximu rearward tilt. The backrest shall be firmly attached to lift carriage. The backrest shall be capable of withstanding without permanent deformation a horizontal static load equal to the rated load uniformly distributed over the area from the top carrying surface of the forks to the top of the backrest for not less than 3 minutes.

3.11 Wheels. Truck shall be a four wheel suspension type. Wheels, when equipped with new tires, shall clear any part of the truck structure and mast structures by at least 1/4-inch under all conditions of operation.

3.12 Tires. The tires shall be new and unused and shall be selected from sizes listed in the TRA Yearbook. The tires shall be of the industrial solid r ber or cushion types molded on to a new press-on type steel band. Front tires shall have nondirectional treads; rear tires may have either smooth or nondirection type treads. Tire loadings shall not exceed the values shown in TRA Yearbook except that in the side handling position, loadings shall not exceed the tire manufacturer's rating for a speed of one (1) mph.

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3.13 Brakes.

3.13.1 Service brakes. The service brakes shall retard, smoothly control, and stop the truck, both with and without load, under all operating conditions specified herein without fade, noise, uneven braking, or malfunction. The brake foot control shall be covered with a replaceable rubber cover. The brake system shall withstand an operator's brake application force of at least 250 pounds at the pedal or treadle without yielding. The service brake system shall be filled with a fluid that is listed on the qualified products list of MIL-B-46176. The orientation of hydraulic wheel cylinders shall permit bleeding of all air in the system through the bleeders. Automatic adjustment of the brakes shall be provided. Automatic adjustment shall maintain correct lining to drum or disc adjustment without looseness or lockup. The automatic adjusters shall be of the guarded or enclosed type, protected from environmental damage or malfunction. The brake system shall conform to the requirements of ANSI B56.1.

3.13.2 Parking brake. A seat-operated parking brake shall be provided. The parking brake shall be an independent mechanical friction brake mounted on the drive motor or drive or propeller shaft rotating with the wheels or a separate mechanical linkage actuating the service brake, capable of holding the truck with rated load on the slope specified for ascension. The brake shall be automatically activated when the operator leaves the operator's seat and hold the loaded vehicle on the specified slope. Adjustment of parking brake shall be accomplished without the removal of any major assembly other than tires and wheels. This adjustment shall provide positive means to prevent loss of adjustment through inadvertent disengagement of adjusting device.

3.14 Body. The truck body (other than counterweight) shall be constructed of steel sheet or plate equal to that used in the manufacturer's normal commercial product. If the horizontal floor plate is more than 26 inches above ground level, an intermediate step shall be provided approximately midway and shall not extend beyond the truck profile. The step shall be large enough to accommodate the foot of the 95th percentile man with an overshoe.

3.15 Seat. A cushion seat and a cushion backrest, adjustable forward and reverse, shall be provided. The seat shall have minimum adjustment, parallel to the longitudinal centerline of the truck, of 4 inches and shall be adjustable by the operator from the operating position without the use of tools. The seat cushion and backrest shall be of foam rubber or polyurethane covered with slip coated vinyl upholstery having a minimum weight of 40 ounces per yard for 54 inch width.

3.16 Wheel guards. If drive or steer wheels, when in a straight-ahead position protrude beyond the body of the truck in excess of 1-1/2 inches, wheel guards shall be provided extending to the outside edge of the tire.

3.17 Towing device. A pintle or pin type towing device shall be provided. It shall be in the center rear of the truck, and shall not protrude beyond the path

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followed by the rear end of the truck when making a turn of minimum turning radius. The towing device shall accept a lunette conforming to MS51336 without binding. The towing device shall be capable of withstanding without permanent deformation a towing force induced in accordance with 4.5.15.

3.18 Slinging and tiedown provisions. Permanently affixed lifting and tiedown provisions that enable the truck to be lifted in its normal travel position and to be fastened to the floor or deck of a transportation medium shall be provided. Provisions shall conform to MIL-STD-209, classes 1 and 2 or class 3 for type II or III equipment, excluding external helicopter transport requirements.

3.19 Pivot and shift assemblies. Pivot and shift assemblies shall be provided to allow 90 degrees pivoting of load and full side shift action to facilitate transfer of pallets (size 40 inches x 48 inches)between the truck and storage position. Pivot and shift shall be accomplished hydraulically. Pivot shall be 90 degrees plus 1 degree minus 5 degrees. With the forks pivoted 90 degrees and shifted to the extreme left or right when transversing to the right or left the transverse motion shall be such that the vertical face of the fork extends 2 inches minimum beyond the right or left side of the truck. If pivot of 180 degrees is furnished, pivot, shift and transverse motion for both direction must be accomplished. With fork in the front handling position, sideshift shall provide for a minimum movement of 12 inches across the width of the truck. With forks in the side handling position and side shifted, the truck shall be capable of carrying 16 inch loads on the deck of the truck within the planned outline of the truck. The pivot and side shift assemblies shall be capable of withstanding tests specified in 4.5.20 and 4.5.21.

3.20 Controls and instrumentation. All controls shall be provided with position marking (decals are not acceptable) either on a diagram visible to the operator or embossed or depressed on or near the control. Load handling controls shall be marked in accordance with ANSI MH 11.3. The distance from the near edge of one control to the near edge of an adjacent control or other surfaces shall be a minimum of 1-1/2 inches. All hydraulic controls shall be conveniently located for right hand operation. Sequence of the control position shall be "LIFT", "TILT", "PIVOT", and "SHIFT". All controls shall provide for feathering of the valve to furnish the metered flow required for precision positioning of the load. Lift and tilt controls shall be self-centering lever(s) and may be combined in a single control lever. Rearward or upward motion of lift and tilt control lever(s) shall raise or tilt load rearward. Pivot and side shifter controls shall be self-centering lever(s) and may be combined in a single control lever. Rearward movement of the side shifter control shall shift the load to the right and forward motion of the lever shall shift the load to the left. Rearward movement of the pivot control shall pivot the load counterclockwise and forward motion of the pivot shall pivot the load clockwise. The pivot and shift control(s) shall be positioned so that the operator can concurrently accomplish pivot and shift motions. Electric controls and instrumentation shall be moisture and weather resistant. Mechanical push-pull remote controls when furnished shall be of the waterproof antifriction

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type, conforming to requirements for marine cables in SAE J917. As a minimum, all exposed parts of the control cable shall be fabricated of corrosion resistant steel. The cables shall be covered with a polyethylene or vinyl jacket. Cable ends shall be tailored to actuate all levers through their respective arcs without binding. Forces necessary to actuate directional and hydraulic controls shall be 10 pounds maximum. All hand-control levers and linkages shall withstand a minimum force of 50 pounds applied at the handgrip in the normal directions of operation and shall have sufficient strength to withstand, without permanent deformation or damage, all vibration forces generated during all operations specified herein. The travel of the controls shall not exceed 6 inches from the center position when measured at the end of the grip.

3.20.1 Instruments. All gages,indicator lights, and meters shall be products of manufacturers regularly engaged in producing these types of instruments. They shall be flush mounted on the instrument panel in view of the operator when he is in operating position. All instruments shall be spark-enclosed construction or mounted in spark-proof enclosures on type EE trucks.

3.20.2 Light switch. A switch for simultaneous operation of the flood-light(s) and taillamp shall be furnished in a location convenient to the operator.

3.20.3 Master switch. A master switch shall be furnished that has provision for a key. When key is removed, control circuits shall be opened or be interrupted making it impossible for any electrical control circuit to be activated. All trucks under each contract shall be keyed identically. Unless otherwise specified (see 6.2), two keys shall be supplied with each truck.

3.20.4 Brake pedal. The brake pedal shall be conveniently located for right foot operation within a clear space at least 6 inches wide.

3.20.5 Accelerator control. The accelerator control shall be installed for comfortable right-foot operation; it shall be located to the right of and shall be free of interference from the brake pedal by a distance of not less than 1-1/2 inches.

3.20.6 Directional controls. Selective forward and reverse control shall be furnished. The control shall be located for convenient left hand operation. Movement of the hand control lever in the forward or upward direction shall cause forward motion of the truck.

3.20.7 Hour meter. An electrically operated hour meter which registers the total number of hours of operation of the traction and pump motors shall be provided. Hour meters shall be in accordance with MIL-M-3971, type I.

3.20.8 Horn. Truck shall be equipped with an electric horn with push button mounted in the center of the steering wheel. Horn button assembly and electrical wiring for the horn shall be constructed to be moisture and weather resistant.

3.20.9 Floodlights. Each truck, shall be equipped with one sealed beam, 25 watt minimum, floodlamp, which shall be shock mounted in an elastomer ring housing on

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th: left side of the mast. Directional adjustment range of this light shall be a mi imum of 45 degrees above and below the horizontal plane. Horizontal adjustment ra.ge shall be a minimum of 300 degrees to illuminate front and rear areas. When specified (see 6.2) trucks shall be equipped with an indentical floodlight mounted on the right side of the mast.

3.20.10 Taillight. One automotive, red, reflector-type, combination stoplight and taillamp shall be recessed or mounted on the rear end within the plan outline of the truck and shall be so located that it will not interfere with the removing or installing of the battery. Bulbs shall be replaceable without removal of battery or counterweight. The stoplight shall operate automatically only upon actuation of the service brake pedal and shall not operate when parking brake is applied.

3.20.11 Electronic guidance control/system. When specified (see 6.2) truck shall be equipped with an electronic wire guidance control system.

3.20.11.1 Electronic guidance control. A sensing device mounted on the truck shall control the forward direction of the truck by detecting a 5 to 10 kilohertz signal, having a signal strength of 250 milliamperes, from a guide wire recessed in the floor. The device shall keep the truck over the wire at all times. A signal level detector shall measure the signal level to permit guidance only when the signal strength is sufficient. The drive motor shall not operate when on guidance without sufficient signal strength. The sensing device shall be capable of receiving the signal from a maximum distance of 4 inches from the guide wire.

3.21 Battery protector. A battery discharge indicator shall be provided and shall be a solid state device. It shall be designed to indicate when the battery is approaching a low state of discharge by first warning the operator by means of a discharge meter, indicating that the battery is approaching 80 percent discharge, followed by an interrupt of the lift control circuit making the lift mechanism inoperative upon reaching 80 percent battery discharge. The circuit shall be such that the operator cannot activate the lift control circuit by unplugging and replugging the battery after it has been 80 percent discharged. There shall be a defined point on the discharge meter where the truck lift mechanism becomes inoperative. Location and method of mounting shall provide plain visibility of the indicator to the driver without creating interference with truck operation, maintenance and entry and egress movements of the operator.

3.22 Driver's overhead guard. An overhead safety guard in accordance with ANSI B56.1 shall be furnished. Guard height, with the mast tilted rearward, measured from the ground level to the top side of guard shall not exceed 83 inches. Guard design shall not interfere with operation of the truck, nor with normal movements of the operator when entering, leaving, or operating the truck. Guard shall not interfere with servicing of the battery and shall permit removal of the battery from the top, in one continuous lift and swing operation. Overhead guard shall remain firmly secured to its support brackets under normal and test course operation. Overhead guard shall completely cover the driver in the operating position.

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3.23 Accessories. Electric accessories and instrumentation shall be moisture and weather resistant when tested in accordance with Section 4.

3.24 Performance.

3.24.1 Capacity. The truck shall have a load lifting capacity of 4000 pounds at 24-inch load center to full lift height when operating as a counter balance front loading unit and 4000 pounds at 24-inch load center to a lift height of 72 inches down rated to 3000 pounds at 24-inch load center to full lift height when operating as a counterbalance front/side loading unit. Except as otherwise specified herein, rated load shall be 4000 pounds.

3.24.2 Maximum fork height. The truck with rated load on the forks shall have a minimum lift height of 194 inches for style 1 truck and 180 inches for style 2 truck, when measured in accordance with 4.5.3.

3.24.3 Lifting speed. Lift speed with rated load shall be not less than 50 feet per minute over the entire distance from ground level to maximum fork height.

3.24.4 Lowering speed. Speed of lowering of unloaded forks shall be not less than 40 feet per minute over the entire distance from maximum fork height to ground level. Speed of lowering forks with rated load shall be not more than 80 feet per minute over the entire distance from maximum fork height to ground level including failure of hydraulic lines.

3.24.5 Steering. The steer wheels shall be capable of being turned from extreme right to extreme left or vice versa in not more than six complete turns of the hand steering wheel. The maximum steering effort through the entire turn shall be not less than 2 pounds nor more than 10 pounds. Free play shall not exceed 3 inches for power steering with the pump motor running. There shall be no permanent damage or deformation of any part of the steering system and related component when subjected to the forces applied in Test Method 3. There shall be a minimum of 7 inches measured vertically between the lowest point on the steering wheel rim and a line parallel to the floor passing through the highest point of the unoccupied seat cushion when seat is held in full down position. There shall be a minumum of 3 inches between rim of steering wheel and directional control lever.

3.24.6 Speed. With forks in the forward position and carrying rated load truck shall be capable of attaining a speed of not less than 6.0 mph on a level surface in a forward and reverse directions.

3.24.7 Upright tilt. When unloaded the truck with forks in either the forward or side carry position shall have a forward tilt of 3 degrees, plus or minus 1/2 degree, and a rearward tilt of 4 degrees minimum.

3.24.8 Slope ascension, forward direction. Truck shall be capable of ascending a 15 percent grade on a dry concrete surface with and without rated load. Truck shall be able to accelerate from a dead stop on this slope when carrying rated load.

3.24.9 Free lift height. Height of forks without increase in the specified collapsed mast height shall be not less than 57 inches.

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3.24.10 Collapsed mast height. The collapsed mast height of unloaded truck measured from the ground to the top of the uprights in a vertical position shall not exceed 89 inches for style 1 truck and 83 inches for style 2 truck.

3.24.11 Underclearance. The truck shall have sufficient underclearance to permit operation from one horizontal plane to another, up and down a 15 percent slope and the obstacle course described in Section 4. Underclearance beneath the mast assembly, when in a true vertical position and subjected to rated load, shall be not less than 2-3/4 inches. When the truck is unloaded, the underclearance of the truck, other than the mast and the axles, shall be a minimum of 4 inches. Hydraulic fittings, hoses, tubing, and linkages shall not be the lowest portion of the truck and shall be protected by structural members from striking obstacles.

3.24.12 Drift. The lift assembly shall be capable of holding the rated load at maximum height not less than 2 minutes with not more than 1-3/4 inch of vertical drift and not more than 1 degree of rotational drift from the vertical. Specified drift rates shall be measured when hydraulic oil temperature at the inlet to each cylinder is not less than 120° F.

3.24.13 Service brakes. A brake pedal pressure of not more than 150 pounds and not less than 50 pounds shall develop a drawbar pull equal to 30 percent of the gross weight of the truck (with rated load) while the truck is being towed over a concrete surface having a coefficient of friction not less than 0.5.

3.24.14 Stability. Truck shall conform to the stability requirements as follows

Operating as Counterbalanced Front Loader		Operating as Counterbalanced Front/Side Loader	
	Platform Slope Value		Platform Slope Value
Test (Stability)	%	Test (Stability)	%
Long. Stacking	4	Long. Stacking	
		(3000 lb Load)	4
Long. Travel	24	(4000 lb Load)	4
Lat. Stacking	6	Long. Travel	24
Lat. Travel	ANSI B56.1	Lat. Stacking (RH)	
		(3000 lb Load)	6
		(4000 lb Load)	6
		Lat. Travel	ANSI B56.1
		Lat. Stacking (LH)	
		(3000 lb Load)	6
		(4000 lb Load)	6

3.24.15 Acceleration. From a standing start the truck, with rated load, shall have an average acceleration rate in the forward direction such as to negotiate a distance of 22 feet in 5 seconds.

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3.24.16 Stopping distance. Truck, with rated load, shall be capable of stopping from a rearward maximum speed in accordance with the applicable requirements of ANSI B56.1 when tested as specified in 4.5.14.

3.24.17 Mechanical parking brake. The parking brake shall be capable of retaining the truck with rated load on a 15 percent slope. Test shall be in accordance with 4.5.6.

3.24.18 Power consumption. The truck with battery shall operate for 6.5 hours when tested in accordance with 4.5.22.

3.24.19 Rain. The truck shall be capable of performing all its normal operations and shall have no accumulation of water in the control box after being subject to the rain test specified in 4.5.23.

3.25 Treatment and painting. All exterior surfaces of the truck shall be thoroughly cleaned and shall be dry and free from mill scale, oil, grease, dirt and rust and shall be painted as soon as practicable after cleaning with not less than one coat of anticorrosive primer and two coats of synthetic enamel. Primer and finish coats shall be applied in accordance with normal commercial practices to assure complete coverage and durability of the finish. The finish color shall be color No. 13538 of FED-STD-595. When a first-produced truck is required for test and evaluation, it shall be painted with one primer coat only. The finish coat, when dry, shall be smooth even surface, free from runs, sags, peels, chips, blisters, and areas of thin film or no film. Surfaces of components and assemblies of the truck not normally painted with a finish coat shall be cleaned and prepared in accordance with good commercial practice. Lift chains, sprockets, rollers, plastics, fabric hoses (except hydraulic hoses) and drive belts shall not be painted except that slight overspray will be permitted provided there is no interference with the proper functioning of the component. Paint must be kept off working surfaces where interference with working of parts would result.

3.26 Walkway coating. Floor plates and step surfaces shall have skid resistant surfaces.

3.27 Fungus and moisture resistance. When specified (see 6.2), the electrical circuitry, including all components and connections except as specified below, shall be protected from the effects of fungus growth and moisture by an overall treatment with a varnish conforming to MIL-V-173, composition I or II as applicable, with 1 percent copper 8-quinolinolate (by weight) based on the nonvolatile content of the varnish:

(a) Components or circuit elements that are inherently fungus and moisture resistant or which are hermetically sealed need not be treated.
(b) Components or circuit elements whose functions will be adversely affected by the varnish coating shall not be treated.

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When used, the varnish shall be applied by spray, brush or a combination of both to give a dry-film thickness of not less than 1 mil to component or element surfaces previously cleaned and prepared so that the surfaces are free from all foreign matter which would interfere with the adherence or function of the varnish.

3.27.1 Composition. Composition II shall be used only in the case when local air pollution regulations in the application of varnish precludes the use of composition I. When composition II is used, the contractor shall provide evidence to the Government that the use of composition II is required, and shall certify that the composition II material complies with Rule 102, Southern California Air Pollution Control District.

3.28 Identification marking. The truck shall be indentified with an identification plate conforming to MIL-P-514, type I, style 1, composition C, of type I, grade A, class 1 material which shall be in a visible and safe location on the bulkheads or panel of the operator's compartment of the truck. The identifying data to be applied to the identification plates shall be as indicated herein and as defined in MIL-STD-130. Identification data shall include National Stock Number (if applicable), contract number, serial number, and U. S. Military Property Number. The truck shall also be equipped with instructions, warning, and caution plates conforming to MIL-P-514, type III, compositon C, of type I, grade A, class 1 material, prominently located and describing any special or important procedure to be followed in operating, loading and servicing of the truck or its components. All plates shall be securely attached to the truck with screws, bolts, or rivets, and shall be furnished and mounted by the contractors. Where plates are required to be attached where the use of fastening devices would reflect performance (such as lift cylinder) adhesives may be used.

3.28.1 Shipping data plate. Shipping data plate shall conform to MIL-P-514, type I, composition C, of type I, grade A, class 1 material and, in addition, shall show the silhouette of the truck in transport position indicating tiedown procedures, the center of gravity and the location and capacity of the lifting attachments. The plates shall be attached by screws, bolts, or rivets in a conspicuous location. Wheel loading information of 3.28.2 may be included on shipping data plate.

3.28.2 Wheel loading plate. Each truck shall be equipped with a wheel loading plate conforming to MIL-P-514, type III, composition C, of type 1, grade A, class 1 material. Plate shall be attached by screws, bolts, or rivets in a conspicuous location. As a minimum a plate shall have the following information:

Wheel loading (no load on forks)	
Drive wheels (each side)	pounds
Steer wheels (each)	pounds
Wheel loading (rated load on forks)	
Drive wheels (each side)	pounds
Steer wheels (each)	pounds

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3.28.3 Body marking. The US Military Property Number shall be indelibly stamped into the external plate of the truck body. The marking shall be located on the right side of the truck, approximately 6 inches below the center of the entrance to the operator's compartment. Marking shall be in 1/4-inch high block letters and numbers.

3.28.4 Truck marking. Markings on truck shall be painted black. Characters shall be block-type capitals and Arabic numerals. Location, information, and character size shall be as follows:

Information	Example	Location	Character Size
Capacity	10,000 lbs Capacity	On each side of mast	2 inch height
Rider warning	NO RIDERS	On rear of mast	2 inch height
Service designation and Registration number	U.S. AIR FORCE 70E XXX	On each side and rear	1-1/2 inch height

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection and test requirements specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections and tests set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Component and material inspection. The contractor is responsible for insuring that components and materials used are manufactured, examined, and tested in accordance with the requirements of referenced specifications and standards.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) First-produced truck inspection (see 4.3).
(b) Quality conformance inspection (see 4.4).
(c) Inspection comparision (see 4.6).
(d) Inspection of packaging (see 4.7).

4.3 First-produced truck inspection. Trucks furnished in accordance with 3.2 shall be examined and tested in accordance with 4.3.1 through 4.3.3. Prior to examination and tests, the following shall be performed;

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(a) Service the truck as required by the manufacturer prior to placing it in service.
(b) Break-in the truck as prescribed by the contractor.
(c) Set the hydraulic system relief pressure as specified by contractor and in 3.9.5 and record. Any change made to this setting shall constitute failure of all tests.
(d) Furnish a schedule of maintenance to be followed during all testing of the first-produced trucks. A list of special tools required to perform this maintenance shall also be provided.

4.3.1 Examination. Prior to testing, the first-produced truck(s) shall be examined for the defects marked "X" in column 1 of table I. Presence of one or more defects shall be cause for rejection.

4.3.2 Tests. Upon successful completion of the examination specified in 4.3.1, the truck(s) and components shall be subjected to the tests marked "X" in column 1 of table II. Acceptance of a first-produced truck shall not exclude the remaining trucks from meeting all the conditions and provisions of this specification and pertinent contract.

4.3.3 Post-test disassembly and examination. Upon successful completion of all tests specified in 4.3.2, the truck shall be subjected to the tests marked "X" in column 3 of table II.

4.3.4 Inspection failure. Failure of a first-produced truck to meet any requirement specified herein during and as a result of the examination and tests specified in 4.3 through 4.3.3 shall be cause for rejection of the first-produced truck(s) and for refusal by the Government to continue acceptance of production trucks until evidence has been provided by the contractor that corrective action has been taken to eliminate the deficiencies. Correction of such deficiencies shall be accomplished by the contractor at no cost to the Government on trucks previously accepted and produced under the contract. Any deficiencies found as a result of the first-produced inspection will be considered prima facie evidence that all trucks accepted prior to the completion of the first-produced inspection are similarly deficient unless evidence to the contrary is furnished by the contractor and such evidence is acceptable to the contracting officer.

4.4 Quality conformance inspection.

4.4.1 Examination. After successful completion of all tests specified in 4.4.2, each truck shall be examined for the defects marked "X" in column 2, of table I. Presence of one or more defects shall be cause for rejection.

4.4.2 Tests. Each truck shall be tested as specified in column 2 of table II. Failure of any test shall be cause for rejection. Hydraulic cylinders may be tested prior to installation in the truck.

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TABLE I. Inspection Schedule.

First-Produced Inspection	Quality Conformance	Defect	Requirement Paragraph
1	2	3	4
X	-	101. Safety not as specified.	3.3
X	X	102. Design not as specified.	3.5
X	X	103. Maintainability not as specified.	3.6
X	X	104. Enclosures and shields not as specified.	3.8.1
X	X	105. Overcurrent devices not as specified.	3.8.2
X	X	106. Static discharge devices not as specified.	3.8.3
X	-	107. Instrument panel not as specified.	3.8.4
X	-	108. Wiring not as specified.	3.8.5
X	X	109. Motors not as specified.	3.9.1.1
X	-	110. Insulation not as specified.	3 9.1.2
X	-	111. Speed controller not as specified.	3.9.1.3
X	X	112. Safety cutout system not as specified.	3.9.1.4
X	X	113. Battery compartment not as specified.	3.9.3
X	-	114. Drive unit not as specified.	3.9.4
X	X	115. Hydraulic system not as specified.	3.9.5
X	X	116. Width not as specified.	3.10.2
X	X	117. Uprights and carriage not as specified.	3.10.4
X	X	118. Fork and fork carriage not as specified.	3.10.4.1
X	X	119. Load backrest not as specified.	3.10.4.2
X	X	120. Wheels not as specified.	3.11
X	-	121. Service brakes not as specified.	3.13.1
X	-	122. Parking brake not as specified.	3.13.2
X	-	123. Truck body not as specified.	3.14
X	X	124. Seat not as specified.	3.15
X	-	125. Wheel guards not as specified.	3.16
X	X	126. Towing device not as specified.	3.17
X	X	127. Slinging and tiedown attachments not as specified.	3.18
X	X	128. Pivot and shift assemblies not as specified.	3.19

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TABLE I. Inspection Schedule. (Continued)

First-Produced Inspection	Quality Conformance	Defect	Requirement Paragraph
1	2	3	4
X	X	129. Controls and instrumentation not as specified.	3.20
X	X	130. Horn not as specified.	3.20.8
X	X	131. Floodlights not as specified.	3.20.9
X	X	132. Taillight not as specified.	3.20.10
X	X	133. Overhead guard not as specified.	3.22
X	X	134. Steering not as specified.	3.24.5
X	X	135. Treatment and painting not as specified.	3.25
X	X	136. Walkway coating not as specified.	3.26
X	X	137. Fungus and moisture resistance not as specified.	3.27
X	X	138. Identification marking not as specified.	3.28
X	X	139. Shipping data plate not as specified.	3.28.1
X	X	140. Wheel loading plate not as specified.	3.28.2
X	X	141. Truck marking not as specified.	3.28.4

TABLE II. Tests.

First Produced	Quality Conformance	Post Test	Test	Test Paragraph	Requirement Paragraph	Test Method No.
1	2	3	4	5	6	7
X	-	X	Collapsed mast height, maximum fork height, free lift height	4.5.3	3.24.10 3.24.2, 3.24.9	1
X	X	X	Lift speed, lowering speed	4.5.4	3.24.3, 3.24.4	2
X	X	X	Travel speed	4.5.5	3.24.6	4
X	X	X	Slope ascension parking brake under clearance	4.5.6	3.24.8, 3.24.17, 3.24.11	5
X	-	X	Tilt	4.5.7	3.24.7	6
X	-	X	Drift	4.5.8	3.24.12	7
X	-	X	Acceleration	4.5.9	3.24.15	8

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TABLE II. Tests. (Continued)

First Produced	Quality Conformance	Post Test	Test	Test Paragraph	Requirement Paragraph	Test Method No.
1	2	3	4	5	6	7
X	-	-	Stability	4.5.10	3.24.14	12
X	X	X	Steering	4.5.11	3.24.5	3
X	-	-	Lifting & tiedown provisions	4.5.12	3.18	-
X	-	-	Mast Assy. Overload	4.5.13	3.10	9
X	X	X	Stopping distance	4.5.14	3.24.16	13
X	-	-	Towing hook	4.5.15	3.17	11
X	-	X	Brakes	4.5.16	3.24.13, 3.13.1	11
X	-	-	Maintainability	4.5.17	3.6, 3.7	10
X	-	-	Fork load backrest	4.5.18	3.10.4.2	18
X	-	-	Tire loading	4.5.19	3.12	14
X	-	-	Side shift	4.5.20	3.19	-
X	-	-	Pivot	4.5.21	3.19	-
X	-	-	Power Consumption	4.5.22	3.24.18	15
X	-	-	Rain protection	4.5.23	3.24.19	16
X	-	-	Hydraulic pump pressure test	4.5.24	3.9.5	-
X	-	-	Hydraulic cylinder	4.5.25	3.9.5.3	20
X	X	-	Hydraulic system cleanliness	4.5.26	3.9.5.4	19
X	-	-	Hydraulic fittings	4.5.27	3.9.5.1	21
X	-	-	Uprights	4.5.28	-	22
X	-	-	Reliability	4.5.29	-	17

4.5 Tests. Tests shall be conducted in accordance with the tests specified in table II. An "X" in the column indicates the tests that shall be conducted.

4.5.1 Test conditions. Unless otherwise specified herein, tests shall be conducted at the ambient temperature and climatic conditions existing at the place of test. Only that maintenance established by the contractor and submitted as maintenance schedule prior to commencement of the tests shall be performed during the tests. Reference test methods are contained in the appendix to this specification.

4.5.2 Test load. During all testing specified herein, test load shall be in accordance with rated load specified in 3.24.1. The tolerances for weight shall be plus 0 percent and minus 1 percent and for dimensions shall be plus or minus 1 percent.

4.5.3. Maximum fork height, collapsed mast height, free-lift height. Test the truck in accordance with Test Method No. 1. Nonconformance to 3.24.10, 3.24.2, and 3.24.9 shall constitute failure of this test.

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4.5.4 Lifting speed and lowering speed. Test the truck in accordance with Test Method No 2. Nonconformance to 3.24.3 or 3.24.4 shall constitute failure of this test.

4.5.5 Travel speed. Test the truck in accordance with Test Method No. 4. Nonconformance to 3.24.6 shall constitute failure of this test.

4.5.6 Slope ascension, parking brake, and underclearance. Test the truck in accordance with Test Method No. 5. Nonconformance to 3.24.8, 3.24.11 or 3.24.17 shall constitute failure of this test.

4.5.7 Tilt. Test the truck in accordance with Test Method No. 6. Nonconformance to 3.24.7 shall constitute failure of this test.

4.5.8 Drift. Test the truck in accordance with Test Method No.7. Nonconformance to 3.24.12 shall constitute failure of this test.

4.5.9 Acceleration. Test the truck in accordance with Test Method No. 8. Nonconformance to 3.24.15 shall constitute failure of this test.

4.5.10 Stability. Conduct "forward stacking", "forward travel", "lateral stacking" and "lateral travel" tilting platform tests in accordance with Test Method No. 12. Inability to attain the stabilities specified in 3.24.14 before overturning shall constitute failure of this test.

4.5.11 Steering. Test the truck in accordance with Test Method No. 3. Nonconformance to 3.24.5 shall constitute failure of this test.

4.5.12 Slinging and tie down attachments. Slinging and tiedown attachments shall be subject to loads specified in MIL-STD-209. Nonconformance to the requirements of MIL-STD-209 shall constitute failure of this test.

4.5.13 Structure overload. Test the truck in accordance with Test Method No. 9. Nonconformance to the requirements of 3.10 shall constitute failure of this test.

4.5.14 Stopping distance. Test the truck in accordance with Test Method No. 13. Nonconformance to the requirements of 3.24.16 shall constitute failure of test.

4.5.15 Towing hook. Test the tow hook in accordance with Test Method No. 11. Nonconformance to the requirements of 3.17 shall constitute failure of this test.

4.5.16 Service brake drag and pressure. Test the truck in accordance with Test Method No. 11. Nonconformance to the requirements of 3.24.13 and 3.13.1 shall constitute failure of this test.

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4.5.17 Maintainability. Test the truck in accordance with Test Method No. 10. Nonconformance to requirements of 3.6 and failure to perform the operation in allotted times as specified in 3.7 shall constitute failure of this test.

4.5.18 Load backrest. Test the backrest in accordance with Test Method No. 18. Nonconformance to 3.10.4.2 shall constitute failure of this test.

4.5.19 Tire loading. Test the truck in accordance with Test Method No. 14. Nonconformance to requirements of 3.12 shall constitute failure of this test.

4.5.20 Sideshift. Operate the sideshift with the forks in the front carry position for 50 full cycles at full speed from one extreme to the other with a 4000 pound load on the forks. Next, operate the sideshift with the forks in the side carry position for 50 full cycles at full speed from one extreme to the other with a 4000 pound load on the forks. During the last 25 cycles of each test, the control shall be released abruptly when the forks are at approximately mid-postion between the left and right extremes and then restarted. Upon completion of the tests, the mast and its components shall be examined for compliance with 3.19. Any evidence of noncompliance shall be cause for rejection of the truck.

4.5.21 Pivot. Operate the pivot for 100 cycles at full speed from one extreme to the other with a 4000 pound load on the forks, during the last 50 cycles, the control shall be released abruptly when the forks are at approximately mid-position between the left and right extremes, and then restarted. Upon completion of the 100 cycles, the mast and its components shall be examined for defects. Nonconformance to the requirements of 3.19 shall constitute failure of this test.

4.5.22 Power consumption. Test the truck in accordance with Test Method No. 15. Nonconformance to the requirements of 3.24.18 shall constitute failure of this test.

4.5.23 Rain protection. Test the truck in accordance with Test Method No. 16. Nonconformance to the requirements of 3.24.19 shall constitute failure of this test.

4.5.24 Hydraulic pump(s). Operate the pump(s) at maximum installed speed and at 1-1/2 times maximum system working pressure for not less than 2 minutes. Check for external leaks at assembled surfaces and the shaft seal. Nonconformance to 3.9.5 shall constitute failure of this test.

4.5.25 Hydraulic cylinder. Test each hydraulic cylinder in accordance with Test Method No. 20. Nonconformance to 3.9.5.3 shall constitute failure of this test.

4.5.26 Hydraulic system cleanliness. Test each truck in accordance with Test Method No. 19. Nonconformance to 3.9.5.4 shall constitute failure of this test.

4.5.27 Hydraulic fittings. Test truck in accordance with Test Method No. 21. Nonconformance to 3.9.5.1 shall constitute failure of this test.

4.5.28 Uprights. Test the uprights in accordance with Test Method No 22. Any evidence of permanent deformation, damage, fracture or inability to complete the required cycles shall constitute failure of this test. Test upright may be other than the upright furnished for reliability test of the first-produced truck.

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4.5.29 Reliability. The first-produced truck(s) shall be tested in accordance with Test Method No. 17. Testing shall be conducted at a rate of 15 to 20 lap /hr. Any failure or evidence of excessive wear shall be cause for rejection of the truck.

4.5.30 Post test, disassembly and inspection. Upon completion of tests marked "X" in column I of table II, the truck shall be subjected to the tests marked "X" in column 3 of table II. Nonconformance to applicable requirements shall constitute a failure. Upon completion of all tests, dismantle the truck sufficiently to permit detailed disassembly and visual inspection of the following components:

(a) Motors.
(b) Differential.
(c) Rear axle bearing and seals.
(d) Front axle bearing and seals.
(e) Complete brake system components.
(f) Steering system cylinders, valves and pump (if used).

Additional disassembly and inspection shall be made at the option of the contracting officer. The disassembly and inspection area shall consist of a paved shop area of sufficient size to permit the disassembly and detailed examination of all truck components. The area should be well lighted, clean, and have provisions for prohibiting all but required personnel. Available equipment shall include tables and benches to permit parts grouping, flashlights, and photographic equipment. Record complete descriptions of the condition of all parts and assemblies examined and take pictures of any deficiency. Each evidence of corrosion which affects the function of the truck, each evidence of permanent deformation, breakage, or excessive wear of any component shall constitute a defect. Manufacturing tolerances listed in the contractor's qualify control program shall be made available upon request and shall be used as the basis for determining excessive wear.

4.5.31 Inspection and test failure. Failure of a first-produced truck to meet any requirement specified herein, during and as a result of the inspection and tests specified in 4.3.1, 4.3.2 and 4.3.3 shall be cause for refusal by the Government to accept any production truck until evidence has been provided by the contractor that corrective action has been taken to eliminate deficiencies. Correction of such deficiencies shall be accomplished by the contractor at no cost to the Government on trucks produced under the contract. Any deficiencies found as a result of the first inspection and tests will be considered prima facie evidence that all trucks accepted prior to the completion of first inspection and tests are similarly deficient unless evidence to the contrary is furnished by the contractor and such evidence is acceptable to the contracting officer.

4.6 Inspection comparison. The Government may select trucks at any time during the contract production period and subject the trucks to the inspections specified in column 1 of table I and to the tests specified in column 1 of table II to determine conformance to the requirements of this specification. The inspection

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will be performed by the Government, at a site selected by the Government, on units selected a random from those which have been accepted by the Government and will not include the previously inspected first-produced truck(s). In addition to any test specified as part of the inspection comparison, the Government reserves the right to conduct any and all other tests contained in this specification as part of the inspection comparison, and failure of such additional tests shall have the same effect as failure of those tests specified as inspection comparison.

4.6.1 Inspection failure. Failure of an inspection comparison truck to meet any requirement specified herein during and as a result of the inspections and tests specified shall be cause for rejection of the inspection comparison truck(s) and shall be cause for refusal by the Government to continue acceptance of production trucks until evidence has been provided by the contractor that corrective action has been taken to elminate the deficiencies. Correction of such deficiencies shall be accomplished by the contractor at no cost to the Government on trucks previously accepted and produced under the contract. Any deficiencies found as a result of the inspection comparison will be considered prima facie evidence that all trucks accepted prior to the completion of inspection comparision are similarly deficient unless evidence to the contrary is furnished by the contractor and such evidence is acceptable to the contracting officer.

4.7 Inspection of packaging. Inspection of the preservation, packing, and marking, shall be in accordance with the quality assurance provisions of MIL-STD-162.

5. PACKAGING

5.1 Preservation, packing, and marking. Preservation, packing and marking shall be level A, level B, level C, or Commercial as specified (see 6.2), in accordance with MIL-STD-162, type I or II as specified (see 6.2).

6. NOTES

6.1 Intended use. Truck is intended to perform in a dual capacity, i.e., to perform all operations normally performed by a conventional 4000-pound capacity, forklift truck to include servicing on-the-road vans and stack penetration in bulk storage areas when operating as a front loader and to perform in narrow aisle rack storage areas when operating as a sideloader.

6.2 Ordering data. Procurement documents shall specify the following:

(a) Title, number, and date of this specification.
(b) Type and style truck to be furnished (see 1.2).
(c) Date of issue of DoDISS applicable to this contract and exceptions thereto (see 2.1.1).
(d) Number of trucks to be furnished. Time frame required for submission of first-produced truck(s) (see 3.2).

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(e) When special tools are not to be provided (see 3.6)
(f) When number of keys other than specified is required (see 3.20.3).
(g) Additional right side mounted light is required (see 3.20.9).
(h) When electronic guidance control system is required (see 3.20.11).
(i) When fungus and moisture resistance is required (see 3.27).
(j) Degree of preservation, and degree of packing required (see 5.1).
(k) Type of packing required (see 5.1).

6.3 First-produced truck(s). Any changes or deviations of production trucks from the approved first-produced model during production will be subject to the approval of the contracting officer. Approval of the first-produced model will not relieve the contractor of his obligation to furnish trucks conforming to this specification.

6.3.1 Incident report. When the contractor conducts the tests specified herein, a verbal or teletype notification within 24 hours and a written report within 48 hours shall be furnished the contracting officer of any incident of equipment malfunction or failure during the conduct of the test. As a minimum, the report shall describe components and parts affected, test and operating conditions, date of incident, hour meter reading, how detected, and description of incident.

6.4 Definitions.

6.4.1 Gross vehicle weight. Gross vehicle weight shall include the complete truck with lubricants and battery necessary for operation and without operator.

6.4.2 Test load. Test load shall be established with a load equivalent to an unrestrained cube with overall dimensions twice the load center dimensions and whose center of gravity is located at the geometric center of the cube. Weight of pallet, if required, shall be included.

6.4.3 Load center. The horizontal longitudinal distance from the intersection of the horizontal load carrying surfaces and vertical load-engaging faces of the forks to the center of gravity of the load.

6.5 Recycled material. It is encouraged that recycled material be used when practical as long as it meets the requirements of the specification (see 3.5.1).

Custodians:
Army - ME
Navy - SA

Preparing activity:
Army - ME

Review activities:
Army - AT, SM, MT
DLA - CS

Project 3930-0497

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APPENDIX

TRUCK, LIFT, FORK, FRONT/SIDE LOADER, ELECTRIC

10. SCOPE

The test methods contained within this appendix are to determine whether trucks procured under this specification conform to the requirements set forth.

20. APPLICABLE DOCUMENTS

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

Recommended Standard Method for Calibration of Liquid Automatic Particle Counters (ANSI B93.28)

(Application for copies should be addressed to the American National Standards Institute, 1430 Broadway, New York, NY 10018.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

Particle Contamination on Surfaces, Measuring and Counting (F24-65)

(Application for copies should be addressed to American Society for Testing and Materials (ASTM) 1916 Race Street, Philadelphia, PA 19103.)

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APPENDIX

TEST METHOD NO. 1

MAXIMUM FORK HEIGHT - FREE LIFT HEIGHT

COLLAPSED MAST HEIGHT

1. Test Course:

   (a) Level, flat surface.

2. Test Apparatus:

   (a) Tape measure.
   (b) Plumb bob.
   (c) Liquid level of clinometer.

3. Test Procedure:

   (a) Measure the true vertical distance from floor to uppermost projection of the collapsed upright assembly with mast in true vertical position as determined by a clinometer. (This is the actual collapsed mast height.)
   (b) Raise rated load until the inner mast begins to exceed the specified collapsed mast height.
   (c) Measure and record the true vertical distance to the top surface of the forks at specified load center. (This is the free lift height.)
   (d) Adjust rated load to 12 inches, plus or minus 4 inches, above ground level and position mast in true vertical position using clinometer and plumb bob.
   (e) Attach the plumb bob to the theoretical intersection of the front vertical surface and top surface of the forks and mark a reference point on the ground.
   (f) Elevate the rated load to the maximum height and check the reference point. If necessary, reposition the mast so that the plumb bob is directly over the original reference point.
   (g) Measure and record the true vertical distance from floor to the top surface of the forks at the specified load center. (This is the maximum fork height.)

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APPENDIX

TEST METHOD NO. 2

LIFTING SPEED - LOWERING SPEED

1. Test Apparatus:

   (a) Tape measure.
   (b) Stopwatch.

2. Test Procedure:

   (a) Measure and record distance in inches from floor to top surface of forks in fully lowered position.
   (b) With battery fully charged record time in secords required to raise rated load to maximum fork height using timing device. Record the difference between maximum fork height and the height of the forks in the lowered position.
   (c) Record time to lower rated load at maximum speed (full open valve).
   (d) Repeat steps (b) and (c) a minimum of 3 times, except no measurements are required.
   (e) Remove load and raise empty forks to maximum lift height.
   (f) Record time in seconds required to lower forks to lowered position.

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APPENDIX

TEST METHOD NO. 3

STEERING

Part I. Wear of Steering Mechanism.*

*This part of test method to be followed before and after reliability test.

1. Test Course: Smooth, clean, dry, and level surfaces.

2. Test Apparatus:

   (a) Piece of stiff fiber board.
   (b) Ruler.

3. Test Procedure:

   (a) Place unloaded truck on a dry, level concrete surface with its front wheels parallel to the longitudinal axis of the truck, with power steering operable.
   (b) Attach to the steering column, immediately below the steering wheel and parallel to it, a piece of stiff fiber board or similar material which is semicircular in shape and which has a radius at least 1 inch greater than the radius of the steering wheel.
   (c) Mark a reference point on the upper portion of the outer edge of the steering wheel; mark a corresponding point on the fiber board templet.
   (d) Turn the steering wheel counterclockwise until the wheels of the truck begin to turn. Mark a point on the fiber board templet corresponding to the reference point on the outer edge of the steering wheel.
   (e) Repeat operation (d) in the clockwise direction.
   (f) Remove the fiber board templet from the truck and place it on a flat surface. Measure the distance between the 2 points made in accordance with operations (d) and (e) above. This distance, in inches, is the free play of the steering wheel.

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APPENDIX

TEST METHOD NO. 3 (CONTINUED)

Part II. Steering Wheel Turning and Tangential Force Tests.

1. Test Course: Smooth, clean, dry, and level surface.

2. Test Apparatus:

(a) Steering wheel adapter pulley.
(b) Tension dynamometer.

3. Test Procedure:

(a) Place unloaded truck on clean, dry, level, smooth concrete with steer wheels in a straight ahead position.
(b) Attach an adapter pully, having a root diameter equal to the steering wheel diameter, to the spokes of the hand steering wheel.
(c) Apply a steady force to a dynamometer attached to the adapter pulley in a counterclockwise direction and in the wheel plane until the wheel turns.
(d) Record the maximum gage reading in 60-degree increments to within 15 degrees of the road wheel steer angle.
(e) Perform (c) and (d) in a clockwise direction.
(f) Steps (b) through (e) may be performed with a torque reading instrument in lieu of an adapter pulley.
(g) Turn steering wheel clockwise until steer wheels hit the stop. Apply 150 pound unbalanced tangential force to the steering wheel and maintain force for a minimum of 15 seconds. Perform this operation two more times.
(h) Repeat step (g) in counterclockwise direction.
(i) Measure the number of turns of the hand steering wheel required to turn the steer wheels from the extreme left position to the opposite direction.
(j) Measure and record steering wheel diameter and steering wheel to seat clearance.

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APPENDIX

TEST METHOD NO. 4

SPEED

1. Test Course:

   (a) Test course shall be a dry, level surface of sufficient length to attain maximum speed prior to entering measured distance, plus measured distance, plus sufficient stopping distance.

2. Test Apparatus:

   (a) Tape measure.
   (b) Stopwatch or electronic timer.

3. Test Procedure:

   (a) Drive the truck a sufficient distance to attain maximum speed prior to entering the measured test course.
   (b) Record length of measured distance, time to traverse measured distance. Measured distance shall be equal to or greater than 44 feet.
   (c) Repeat for a total of six runs, three in each direction.
   (d) Repeat total of six runs in reverse gear, three in each direction, except only time measurement is required. Travel speeds in forward and reverse shall be the average of the six runs in the corresponding runs.

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APPENDIX

TEST METHOD NO. 5

SLOPE ASCENSION - PARKING BRAKE - UNDERCLEARANCE

1. Test Course:

   (a) Tilt table or ramp.
   (b) Underclearance template (optional).

2. Test Apparatus:

   (a) Clinometer.

3. Test Procedure.

   (a) Drive truck forward with rated load along horizontal approach and onto ramp of specified percent grade and bring to complete stop. (All wheels supported on the ramp.)
   (b) Apply parking brake.
   (c) Release service brakes and observe whether truck remains stationary.
   (d) Release parking brake and start from a standstill. Proceed up ramp and onto a horizontal surface. Observe underclearance.
   (e) Repeat steps 3(a) through 3(d) without load.
   (f) Repeat steps 3(a) through 3(c) with truck in reverse.
   (g) In lieu of driving onto a horizontal surface at top of incline, a template may be used to check underclearance.
   (h) Measure and record underclearance beneath the mast assembly, beneath each axle and the frame of the truck. Record the underclearance of the lowest component and note the component. Note whether any hydraulic components and linkages are not protected.

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APPENDIX

TEST METHOD NO. 6

UPRIGHT TILT

1. Test Course:

   (a) Level, flat surface.

2. Test Apparatus:

   (a) Clinometer.

3. Test Procedure:

   (a) Tilt.

      (1) Place truck without load on flat level surface, using clinometer to insure truck is level.
      (2) Raise forks to an elevation of 2 feet above the surface, tilt mast as far forward as possible, and record angle of tilt shown on clinometer located on front surface of outer mast channel.
      (3) Tilt mast as far rearward as possible and record angle of tilt on clinometer located on front surface of outer mast channel.

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APPENDIX

TEST METHOD NO. 7

DRIFT

1. Test Course:

   (a) Level, flat surface.

2. Test Apparatus:

   (a) Temperature thermocouples.
   (b) Clinometer.

3. Test Procedure:

   (a) Install thermocouple in the hydraulic reservoir. Heat the hydraulic oil by exercising the lift and tilt functions and maintain hydraulic oil temperature above 120° F temperature during the duration of the test.
   (b) Raise rated load to maximum fork height obtainable with outer channel in true vertical position measured with the clinometer. Mark reference point on mast upper stage.
   (c) Record initial outer mast tilt and hydraulic oil temperatures.
   (d) Hold for specified time and record downward drift, rotational drift, and hydraulic temperature.

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APPENDIX

TEST METHOD NO. 8

ACCELERATION

1. Test Course:

   (a) Level, flat surface.

2. Test Apparatus:

   (a) Tape measure.
   (b) Stopwatch or electronic timer.

3. Test Procedure:

   (a) Carrying a 4000 pound load in the front loading position accelerate the truck in forward drive at maximum speed through the specified distance from a standing start.
   (b) Repeat (a) for a total of six runs, three in each direction, except only time measurement is required.
   (c) Acceleration time shall be the average of six runs.

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APPENDIX

TEST METHOD NO. 9

STRUCTURE OVERLOAD

1. Test Course:

(a) Level, flat surface.
(b) Anchor.

2. Test Apparatus:

(a) Magnetic particle or dye penetrant capability.

3. Test Procedure:

(a) Front loading position.

(1) Place truck on level surface and support it with blocks under the axle or frame to relieve strain on tires. Position mast in true vertical position, with a 4000 pound load on the forks in the front carrying position. External means may be utilized to hydraulically lock the tilt cylinder.
(2) Secure truck to floor in a manner to compensate for the additional test loads to be placed on forks. Counterweight may be added to assist the holding force. Apply counter-balancing forces in a vertical direction.
(3) With crane or another truck, place a test load of the rated capacity in position over the first test load such that the center of mass of the test load is on the truck's longitudinal centerline and at the rated load center.
(4) Check strain or holddown devices and hydraulic system to determine whether the additional test load can be placed on forks. Tighten holddown devices if necessary.
(5) With the crane or another truck, place the third test load carefully in position over the first two loads so that the center of mass of load is on the longitudinal centerline of the truck and at the rated load center.
(6) When placing the overload on trucks, personnel should stand clear and crane operator should stand by to remove load if there is any evidence of immediate collapse or failure in the truck structure.
(7) The 300 percent rated load may be applied as one load in lieu of three loads as described in (3) through (6).
(8) If truck accepts the additional test load without immediate failure, let overload stand on truck for 10 minutes.
(9) Remove loads.

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APPENDIX

TEST METHOD NO. 9 (CONT'D)

(10) Inspect truck structure, frame, mast assembly, and the hydraulic system for evidence of deformation, fractures, leakage in hydraulic system, broken welds, etc.
(11) Inspect forks a minimum of 6 inches each side of heel by magnetic particle of dye penetrant method.

(b) Side loading position

(1) Test may be conducted with the identical truck used in (a) or a second truck may be substituted for this test. Place truck on level surface and support it with blocks under the axle or frame to relieve strain on tires. With a 4000 pound load on the forks in the front loading position, pivot the forks 90-degrees and shift the forks to the side loading position. Mast should be positioned in the true vertical position. External means may be utilized to hydraulically lock the tilt cylinder. Repeat steps 2 thru 11 above.

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APPENDIX

TEST METHOD NO. 10

MAINTAINABILITY

1. Test Course:

Shop area of sufficient size to permit unobstructed surveillance of maintenance.

2. Test Apparatus:

(a) Common handtools.
(b) Special tools (if furnished with truck).

3. Test Procedure:

(a) Examine truck for accessibility of major assemblies.
(b) Examine position of drains with respect to accessibility.
(c) Using the tools, demonstrate that major assemblies are accessible for repair and maintenance.
(d) Demonstrate both the accessibility of drains and the path of discharge of lubricants by activating drainage controls.
(e) Perform and time each operation on the maintenance operations list (see 3.7).

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APPENDIX

TEST METHOD NO. 11

SERVICE BRAKE, DRAG AND PRESSURE, TOWING HOOK STRENGTH

1. Test Course:

Level, clean, dry, concrete, having a coefficient of friction not less than .50.

2. Test Apparatus:

(a) Pedal force gage.
(b) Tension dynamometer.
(c) Block and tackle.

3. Test Procedure:

A. Brake Pressure Test.

(a) Attach the pressure gage to the brake pedal in a manner which enables pressure to be applied to the face of the gage.
(b) Apply 250 pound pressure to the brake pedal, hold for 1 minute and release. Repeat this operation 10 times.
(c) Inspect for component failure.

B. Brake Drag and Towing Hook Strength.

(a) Determine gross weight of truck with rated load.
(b) Attach pressure gage to brake pedal in a manner which will enable pressure to be applied to the face of the gage.
(c) Attach a tension dynamometer to the front of the truck with mast in maximum rearward tilt.
(d) Attach one end of a block and tackle, or similar mechanical device, to the tension dynamometer, and the other end to an anchoring device.
(e) With operator seated in operator's seat, disengage handbrake and place directional control lever in neutral.
(f) Apply the following brake pedal forces and record the dynamometer reading:

(1) 50 lbs brake pedal force.
(2) 150 lbs brake pedal force.

(g) Repeat operations (b) through (f) with dynamometer attached to rear towing eye.
(h) Examine towing eye for deformation.

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APPENDIX

TEST METHOD NO. 12

LONGITUDINAL AND LATERAL STABILITY

1. Test Course:

(a) Tilt table.

2. Test Apparatus:

(a) Clinometer
(b) Plumb bob
(c) Tape measure

3. Test Procedure:

A Counter Balanced Front Loading Operation

(a) Longitudinal Stability - Stacking

(1) Place the truck on an initially level platform with the axis of the load wheels parallel to the tilting axis of the platform. (Ref. Figure A-1).
(2) When placing the rated load on the forks, the center of mass of the load shall be on the longitudinal centerline of the mast which shall be on the longitudinal centerline of the truck. For trucks with tilting uprights with the load supported by the forks approximately 12 inches (300 mm) above the platform, and in a normal and unrestrained manner with the stationary mast being vertical with the level platform, attach a plumb line at the theoretical intersection of the vertical face and top surfaces of the forks to establish a reference point. (A transit or other methods may be used instead of a plumb line to establish vertical.)
(3) With the rated load elevated to maximum height, and for trucks with tilting upright, the test platform shall be tipped to a 4 percent slope. During the test there shall be no operator on the truck. Maintain the truck position on the platform by brakes or other similar means, but not by means of wheel chocks.

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APPENDIX

TEST METHOD NO.12 (CONT'D)

(b) Longitudinal Stability - Traveling

(1) Place the truck on an intitially level platform with the axis of the load wheels parallel to the tilting axis of the platform. (Ref. Figure A-1).
(2) When placing the rated load on the forks, the center of mass of the load shall be on the longitudinal centerline of the mast which shall be on the longitudinal centerline of the truck. The mast shall be at full rearward tilt. The load is then elevated until the point of intersection of the front vertical face and top surface of the forks is 12 inches (300 mm) from the floor.
(3) With the load elevated as described, tip the test platform to a 24 percent slope. During the test there shall be no operator on the truck. Maintain the truck position on the platform by brakes or other similar means, but not by wheel chocks.

(c) Lateral Stability - Stacking

(1) Placement of Truck on Tilt Platform

a. Place truck, with an articulated steer axle, on an intitally level platform in a position that a line connecting the center of the right load wheel tire (or outermost tire where multiple tires are used) and the projection perpendicular to the platform of the point of intersection of the longitudinal centerline of the truck and the centerline of the steering wheels is paralled to the axis of tilting platform. Steering wheels shall be positioned parallel to the tilting axis of the platform (see Figure A-2).
b. Place truck, with a non-articulated steer axle, on an initially level platform in a position that a line connecting the center of right load wheel tire (or outermost tire where multiple tires are used) and the projection perpendicular to the platform of the point of intersection of the respective steer tire centerline is parallel to the axis of the tilting platform. Steer wheels shall be positioned such that they are parallel to the tilting axis of the platform (see Figure A-3).

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APPENDIX

TEST METHOD NO. 12 (CONT'D)

(2) Place the rated load on the forks, the center of mass of the load shall be on the longitudinal centerline of the mast. Shift the mast and/or the load to the position of least lateral stability. Elevate the load approximately 12 inches (300 mm) above the platform and position mast in true vertical position using level and plumb bob, or equivalent, to a marked reference point on the platform.

(3) Elevate the load to maximum height, and reposition mast to the reference point as required. The test platform shall be tipped to a 6 percent slope. During the test there shall be no operator on the truck. Maintain the truck position on the platform by brakes or other similar means, but not by means of wheel chocks.

(4) Repeat a. and b. using left load wheel in lieu of right load wheel.

(e) Lateral Stability - Traveling (No Load)

(1) Placement of Truck on Tilt Platform

a. Place empty truck with a articulated steer axle, on an initially level platform in a position that a line connecting the center of the right load wheel tire (or outermost tire where multiple tires are used) and the projection perpendicular to the platform of the point of intersection of the longitudinal centerline of the truck and the centerline of the steering wheels is parallel to the axis of the tilting platform. Steering wheels shall be positioned paralled to the tilting axis of the platform (see Figure A-4).

b. Place empty truck, with a non-articulated steer axle, on an initially level platform in a positon that a line connecting the center of the right load wheel tire (or outermost tire where multiple tires are used) and the projection perpendicular to the platform of the point of intersection of the respective steer tire centerline is parallel to the axis of the tilting platform. Steering wheels shall be positioned parallel to the tilting axis of the platform (see Figure A-5).

c. Position the steering wheel nearest the tilting axis parallel to the tilting axis of the platform.

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APPENDIX

TEST METHOD NO. 12 (CONT'D)

(2) Position the mast to full rearward tilt and elevate the forks until the point of intersection of the front vertical face and top surfaces of the forks are 12 inches (300 mm) from the floor.
(3) With the forks elevated as described, tip the test platform to the minimum slope specified in ANSI B56.1. During the test there shall be no operator on the truck. Maintain the truck position on the platform by brakes or other similar means, but not by means of wheel chocks.
(4) Repeat a. and b. using left load wheel in lieu of right load wheel.

B Counterbalanced Front/Side Loading Operation

(a) Longitudinal Stability - Stacking

(1) Place the truck on an initially level platform with the axis of the load wheels parallel to the tilting axis of the platform (Ref. Figure A-6).
(2) Place a 3000 pound load on the forks with the center of mass of the load on the longitudinal centerline of the mast with the forks in the front carry mode. The load shall be pivoted to the position of least longitudinal stability. Elevate the load approximately 12 inches (300 mm) above platform and position mast in true vertical position using level and plumb bob, or equivalent to a marked reference point on the platform.
(3) With the load elevated to maximum height and with mast repositioned to the reference point as required the test platform shall be tipped to a 4 percent slope. During the test there shall be no operation on the truck. Maintain the truck position on the platform by brakes or other similar means, but not by means of wheel chocks.
(4) Now, place a 4000 pound load on the forks with the center of mass of the load on the longitudinal centerline of the mast with the forks in the front carry mode. The load shall be pivoted to the position of least longitudinal stability. Elevate the load approximately 12 inches (300 mm) above platform and position mast in true vertical position using level and plumb bob, or equivalent.

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APPENDIX

TEST METHOD NO. 12 (CONT'D)

(5) With the load elevated to 72 inch (minimum) height, and with mast repositioned to the reference point as required the test platform shall be tipped to a 4 percent slope. During the test there shall be no operation on the truck. Maintain the truck position on the platform by brakes or other similar means, but not by means of wheel chocks.

(b) Longitudinal Stability - Traveling

(1) Place the truck on an initially level platform with the axis of the load wheels parallel to the tilting axis of the platform. (Ref. Figure A-7).
(2) Place a 4000 pound load on the forks, and pivot the load 90° to either the right or left. Position load such that the center of mass of the load is on the intersection of the centerline of the truck and mast. Position the mast at full rearward tilt. Elevate load to a position where the intersection of the vertical front face and top surface of the forks are 12 inches (300 mm) from the floor.
(3) Tip the test platform to a 24 percent slope. During the test there should be no operator on the truck. Maintain the truck position on the platform by brakes or other similar means, but not by wheel chocks.

(c) Lateral Stability - Stacking (Right Hand)

(1) Placement of Truck

a. Place truck, with an articulated steer axle, on an initially level platform in a position that a line connecting the center of the right load wheel tire (or outermost tire where multiple tires are used) and the projection perpendicular to the platform of the point of intersection of the longitudinal centerline of the truck and the centerline of the steering wheels is parallel to the axis of tilting platform. Steering wheels shall be positioned such that they are parallel to the tilting axis of the platform (see Figure A-8).
b. Place truck, with a non-articulated steer axle, on an initially level platform in a position that a line connecting the center of the right load wheel tire (or outermost tire where multiple tires are used) and the projection perpendicular to the platform of

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the point intersection of the respective steer tire centerline is parallel to the axis of the tilting platform. Steering wheels shall be positioned such that they are parallel to the tilting axis of the platform (see Figure A-9).

(2) Place a 3000 pound load on the forks with the center of mass of the load on the longitudinal centerline of the mast which shall be on the longitudinal centerline of the truck with the forks in the front carry mode. Pivot and side shift the load to the position of least lateral stability. Elevate the load approximately 12 inches (300 mm) above the platform and position mast in true vertical position using level and plumb bob, or equivalent.

(3) With the load elevated to maximum height, and the mast repositioned to true vertical, the test platform shall be tipped to a 6 percent slope. During the test there should be no operator on the truck. Maintain the truck position on the platform by brakes or other similar means, but not by means of wheel chocks.

(4) Now, place a 4000 pound load on the forks with the center of mass of the load on the longitudinal centerline of the mast which shall be on the longitudinal centerline of the truck with the forks in the front carry mode. Pivot the load to the position of least lateral stability. Elevate the load approximately 12 inches (300 mm) above the platform and position mast in true vertical position using level and plumb bob, or equivalent.

(5) With the load elevated to 72-inch (minimum) height, and mast repositioned to true vertical, the test platform shall be tipped to a 6 percent slope. During the test there shall be no operation on the truck. Maintain the truck position on the platform by brakes or other similar means, but not by means of wheel chocks.

(6) Repeat test (c) using left load wheel in lieu of right load wheel.

(d) Lateral Stability - Traveling

(1) Placement of Truck

a. Place empty truck with an articulated steer axle, on an initially level platform in a position that a line connecting the center of the left load wheel tire (or

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outermost tire where multiple tires are used) and the projection perpendicular to platform of the point of intersection of the longitudinal centerline of the truck and the centerline of the steering wheels is parallel to the axis of the tilting of the platform. Steering wheels shall be positioned such that they are parallel to the tilt axis of the platform (see Figure A-10).

b. Place empty truck, with a non-articulated steer axle, on an initially level platform in a position that a line connecting the center of the left load wheel tire (or outermost tire where multiple tires are used) and the projection perpendicular to the platform of the point of intersection of the respective steer tire centerline is parallel to the axis of the tilting platform. Steering wheels shall be positioned such that they are parallel to the tilting axis of the platform (see Figure A-11).

(2) Place a 4000 pound load on the forks and pivot the load 90 degrees from the longitudinal centerline of the chassis and shift to the position of least stability. If the truck has a tilting mast, the mast shall be either vertical or at full rearward tilt, whichever is less stable. Elevate the forks until the point of intersection of the front vertical face and top surface of the forks are 12 inches (300 mm) from the floor.

(3) Tip the platform to the minimum slope specified in ANSI B56.1 During the test there shall be no operator on the truck. Maintain the truck position on the platform by brakes or other similar means, but not by means of wheel chocks.

(4) Repeat test using right load wheel in lieu of left load wheel.

(e) Lateral Stability - Stacking (Left Hand)

(1) Placement of Truck

a. Place empty truck with an articulated steer axle, on an initially level platform in a position that a line connecting the center of the left load wheel tire (or outermost tire where multiple tires are

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APPENDIX

TEST METHOD NO 12 (CONT'D)

used) and the projection perpendicular to platform of the point of intersection of the longitudinal centerline of the truck and the centerline of the steering wheels is parallel to the axis of the tilting of the platform. Position steering wheels parallel to the tilting axis of the platform (see Figure A-12).

b. Place empty truck, with a non-articulated steer axle, on an initially level platform in a position that a line connecting the center of the left load wheel tire (or outermost tire where multiple tires are used) and the projection perpendicular to the platform of the point of intersection of the

respective steer tire centerline is parallel to the axis of the tilting platform. Position steering wheels parallel to the tilting axis of the platform (see Figure A-13).

(2) Place a 3000 pound load on the forks with the center of mass of the load on the longitudinal centerline of the mast with the forks in the front carry mode. The load shall be tilted toward the platform tilt axis, shifted and pivoted to the position of least stability.

(3) With the load elevated to maximum height, the test platform should be tipped to a 6 percent slope. During the test there shall be no operator on the truck. Maintain the truck position on the platform by brakes or similar means, but not by means of wheel chocks.

(4) Now, place a 4000 pound load on the forks with the center of mass of the load on the longitudinal centerline of the mast with the forks in the front carry mode. The load shall be tilted toward the platform tilt axis, shifted and pivoted to the position of least stability.

(5) With load elevated to 72-inch (minimum) height, the test platform shall be tipped to a 6 percent slope. During the test there shall be no operator on the truck. Maintain the truck position on the platform by brakes or other means, but not by means of wheel chocks.

(6) Repeat test using right load wheel in lieu of left.

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APPENDIX

IN ALL FIGURES (A-1 THROUGH A-12)

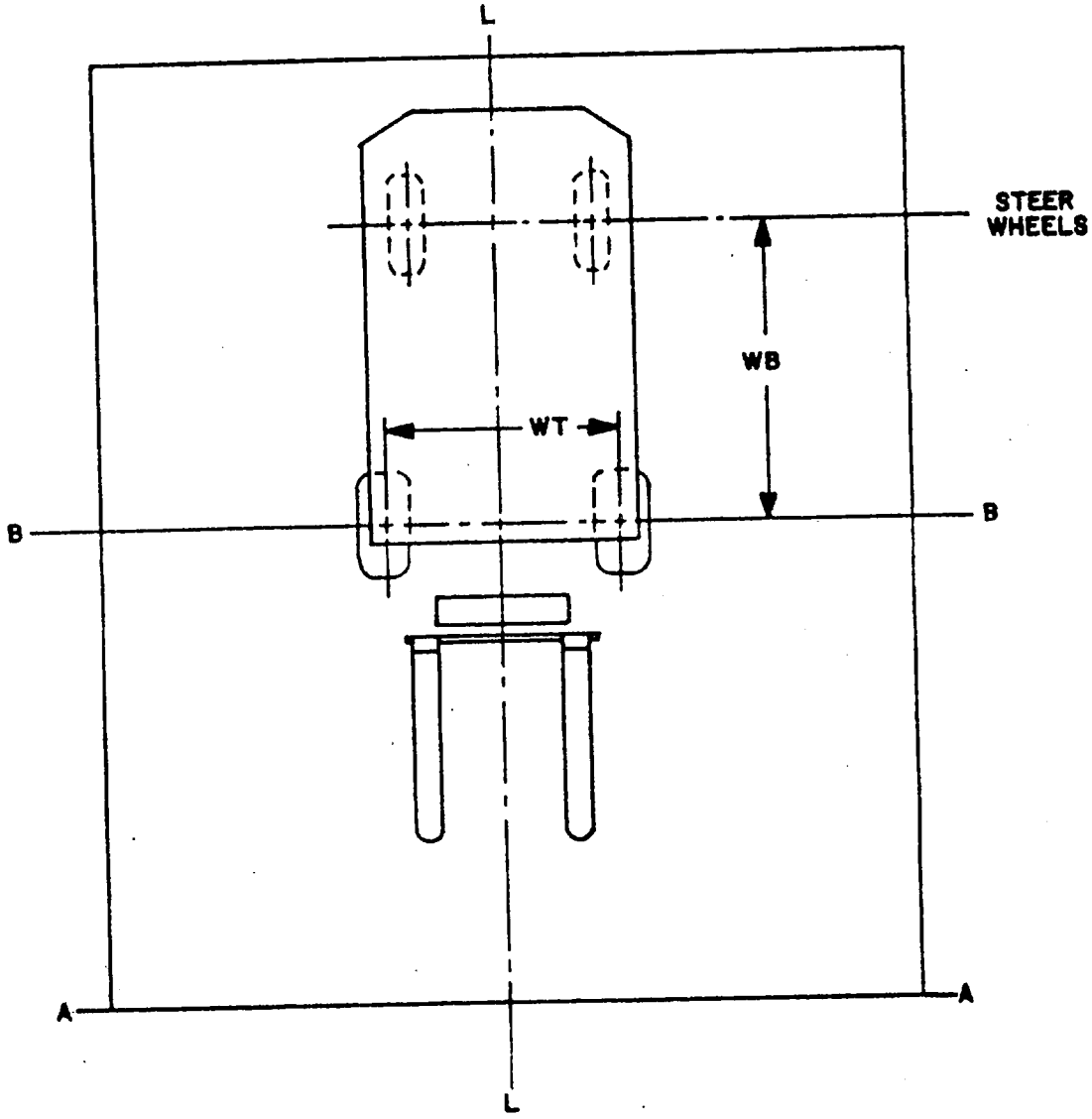
CONCERNING STABILITY TESTING

The following identifications apply:

A-A = Test Platform Tilt Axis
B-B = Axis about Which Truck Upsets
C-C = Parallel to Tilting Axis of Test Platform
L-L = Truck Longitudinal Centerline
I = Points of Intersection
WT = Wheel Tread
WB = Wheelbase

Tilting platform is tilted "up" about axis by raising opposite edge of platform.

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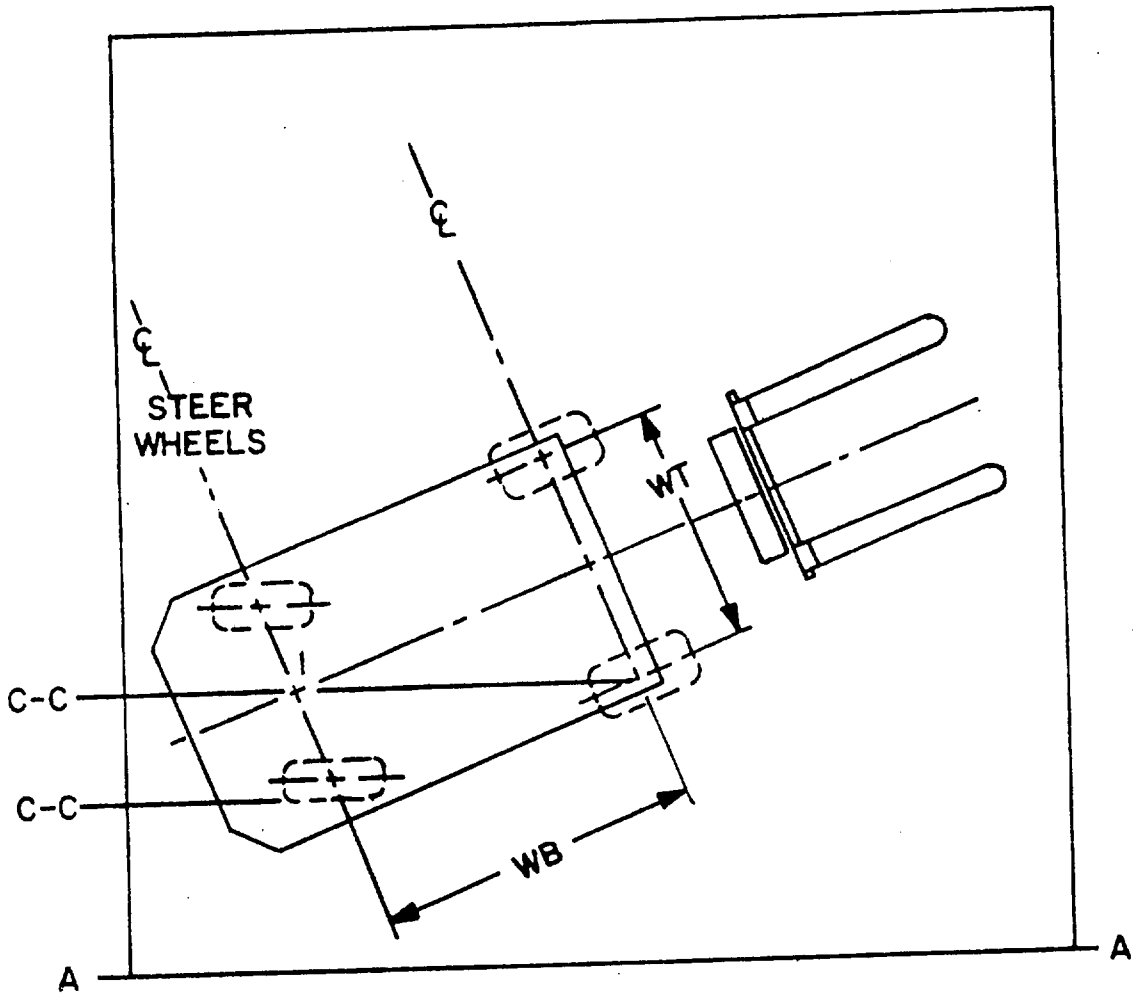


X-3983A

FIGURE A-1. Longitudinal stability

(Counterbalanced front loading operation.)

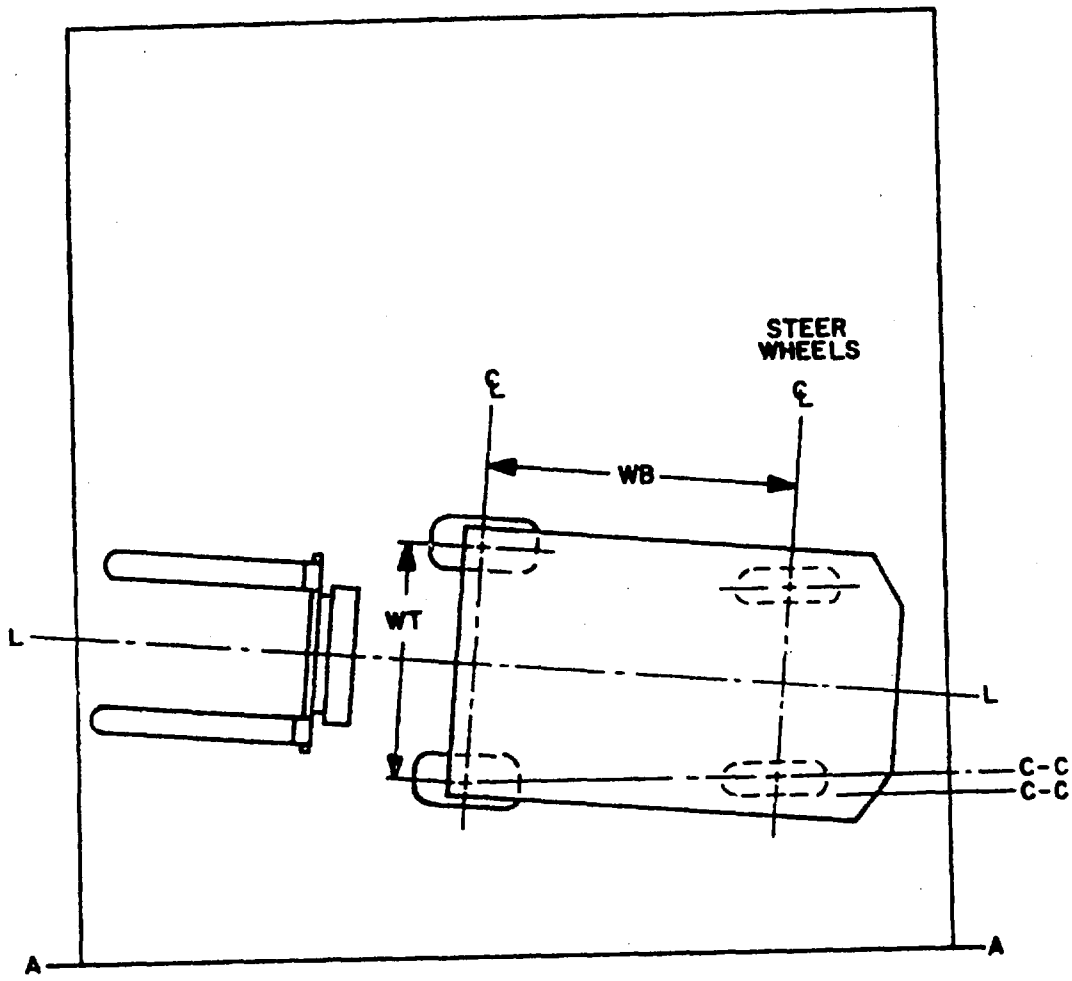
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NOTE:
RIGHT HAND POSITION SHOWN, LEFT HAND POSITION, OPPOSITE.

FIGURE A-2. Lateral stability-stacking. X-3984A
(articulated steer axle)
(counter balance front loading operation).

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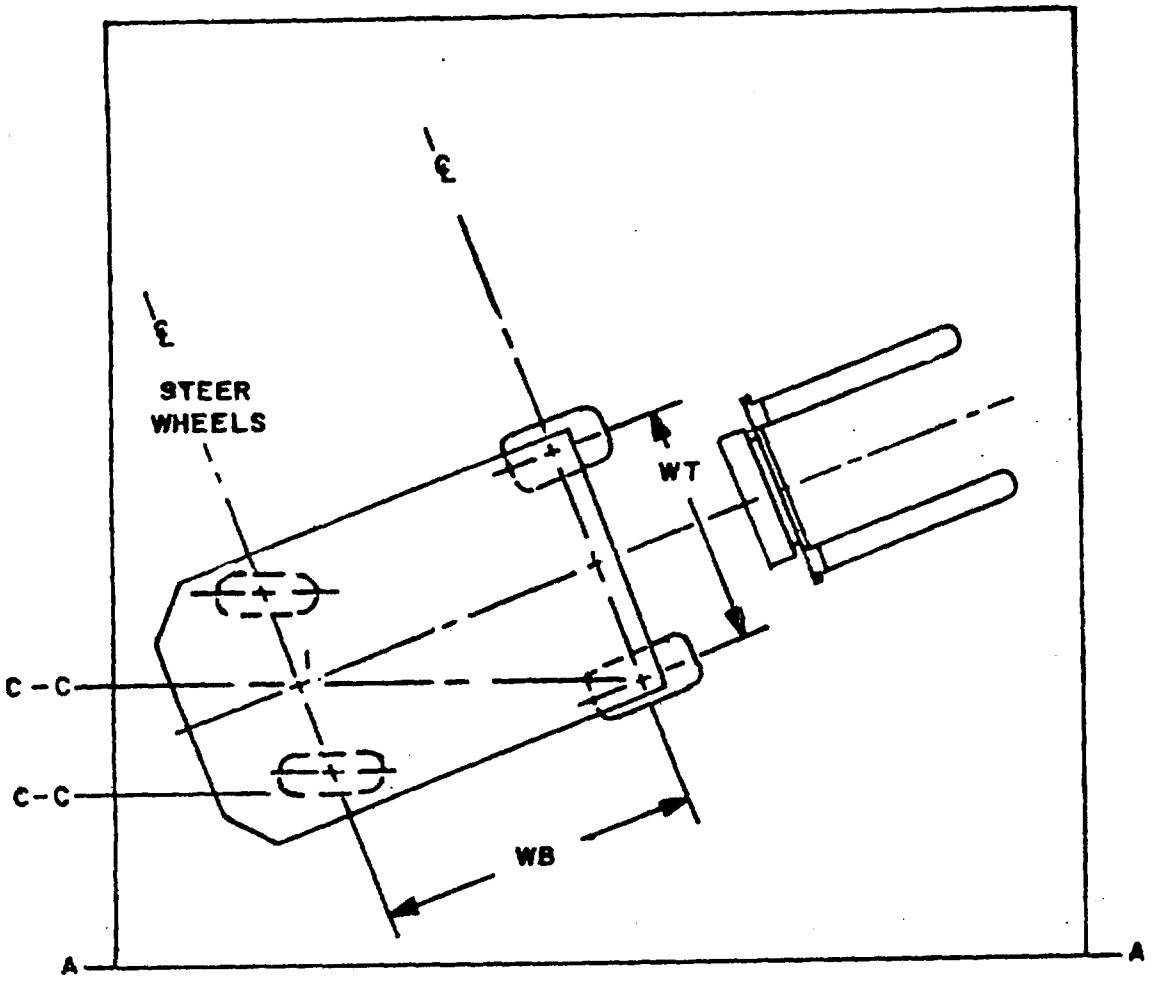
NOTE:

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FIGURE A-3. Lateral stability-stacking (non-articulated steer axle) (counterbalanced front loading operation).

X-3985A

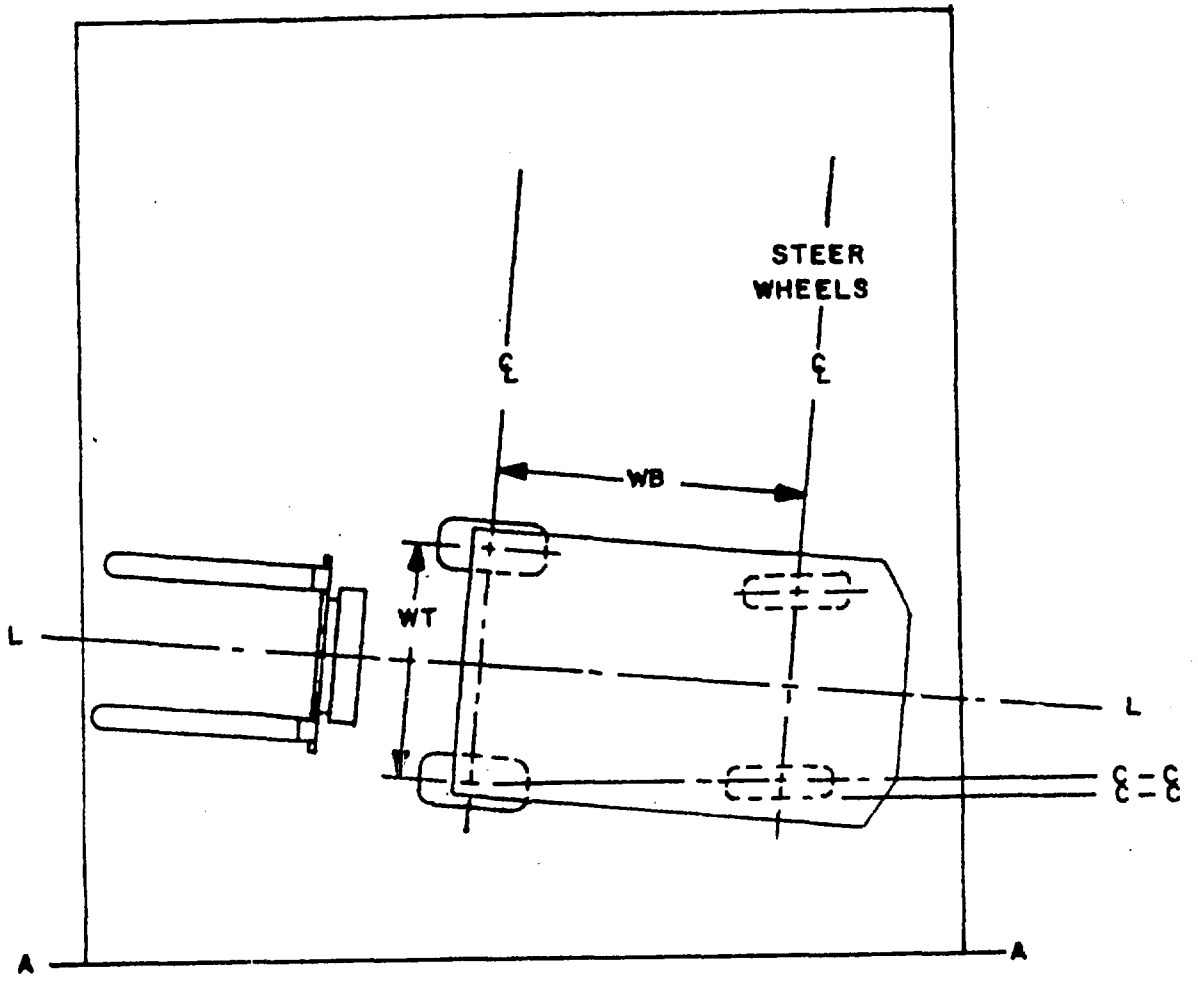
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NOTE:
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FIGURE A-4. Lateral stability-traveling. X-3986A
(articulated steer axle)
(counterbalanced front loading operation).

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NOTE:
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FIGURE A-5. Lateral stability - traveling. X-3987A
(non-articulated steer axle)
(counterbalanced front loading operation).

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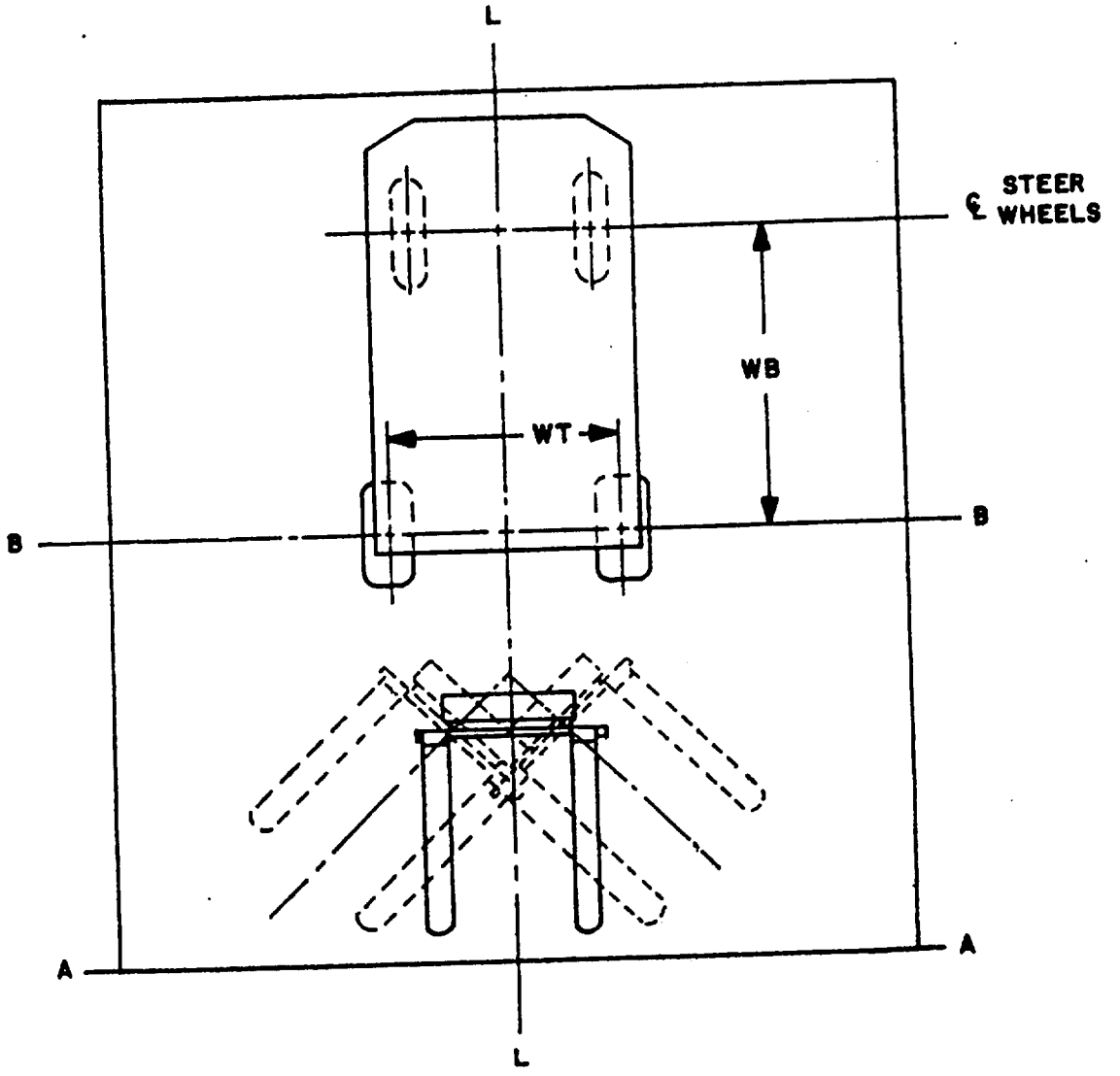


FIGURE A-6. Longitudinal stability – stacking.
(Counterbalanced front/side loading operation.)

X-3988A

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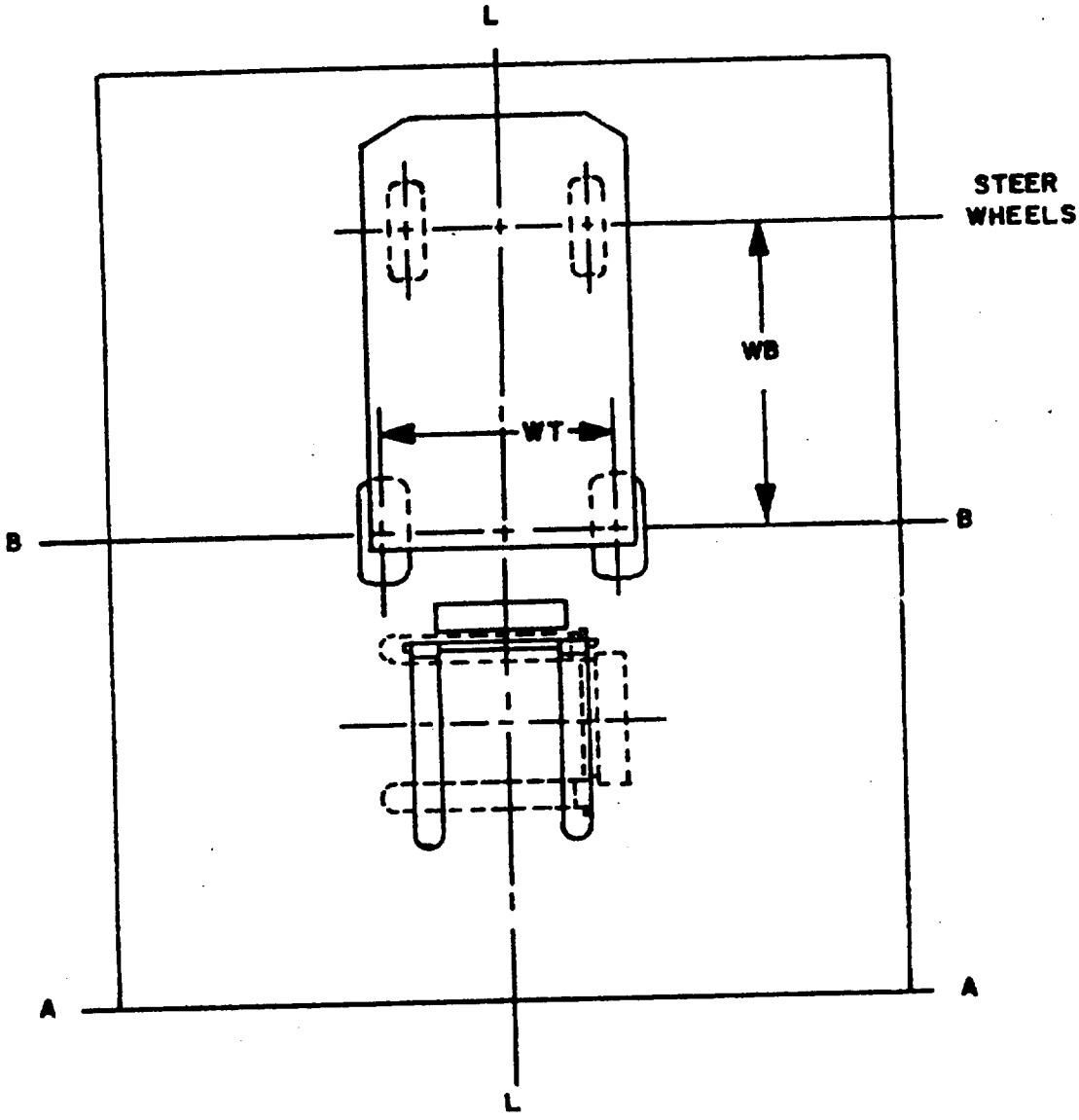
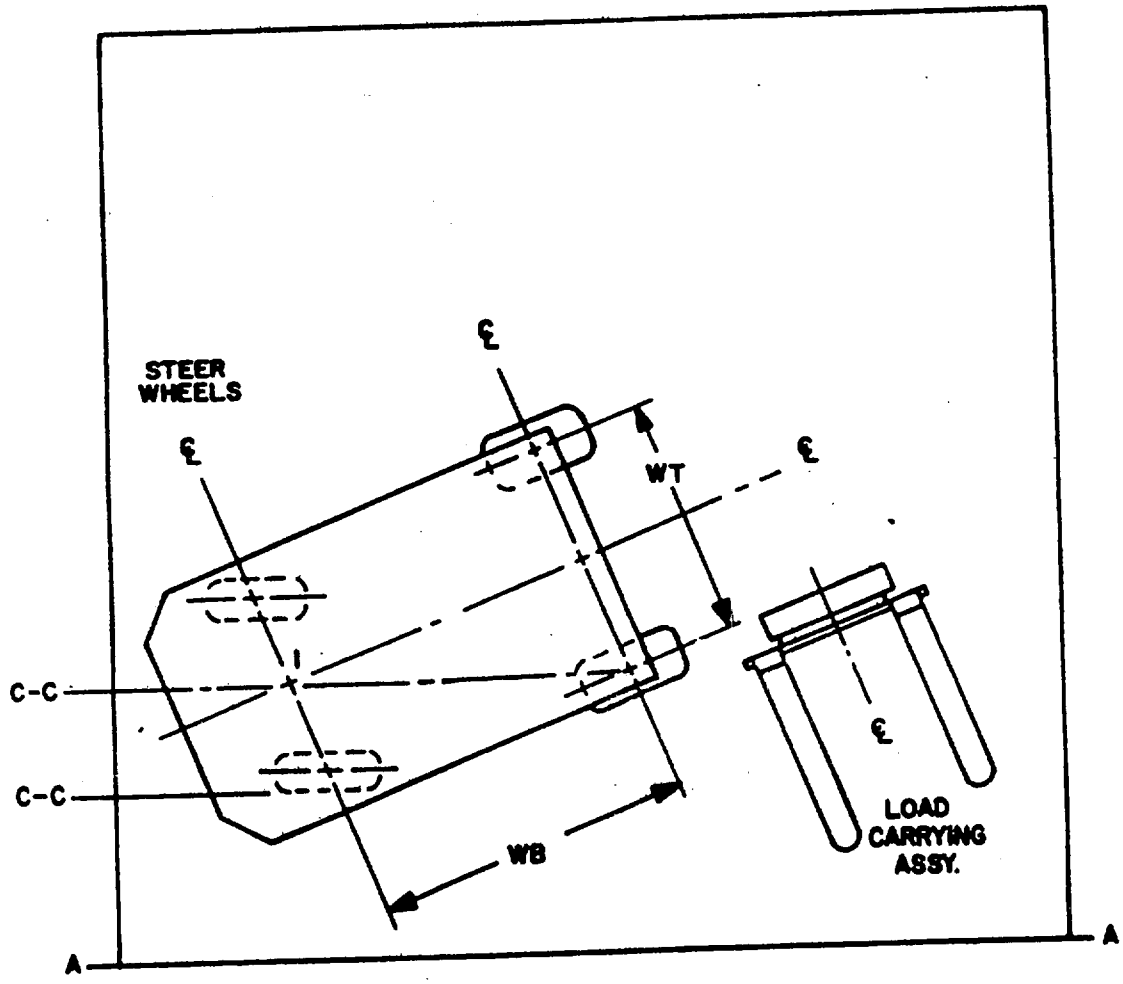


FIGURE A-7. Longitudinal stability-traveling. X-3989A

(Counterbalanced front / side loading operation.)

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NOTE:
RIGHT HAND POSITION SHOWN, LEFT HAND POSITION, OPPOSITE.

X-3990A

FIGURE A-8 Lateral stability-stacking. (articulated steer axle) (counterbalanced front/side loading operation).

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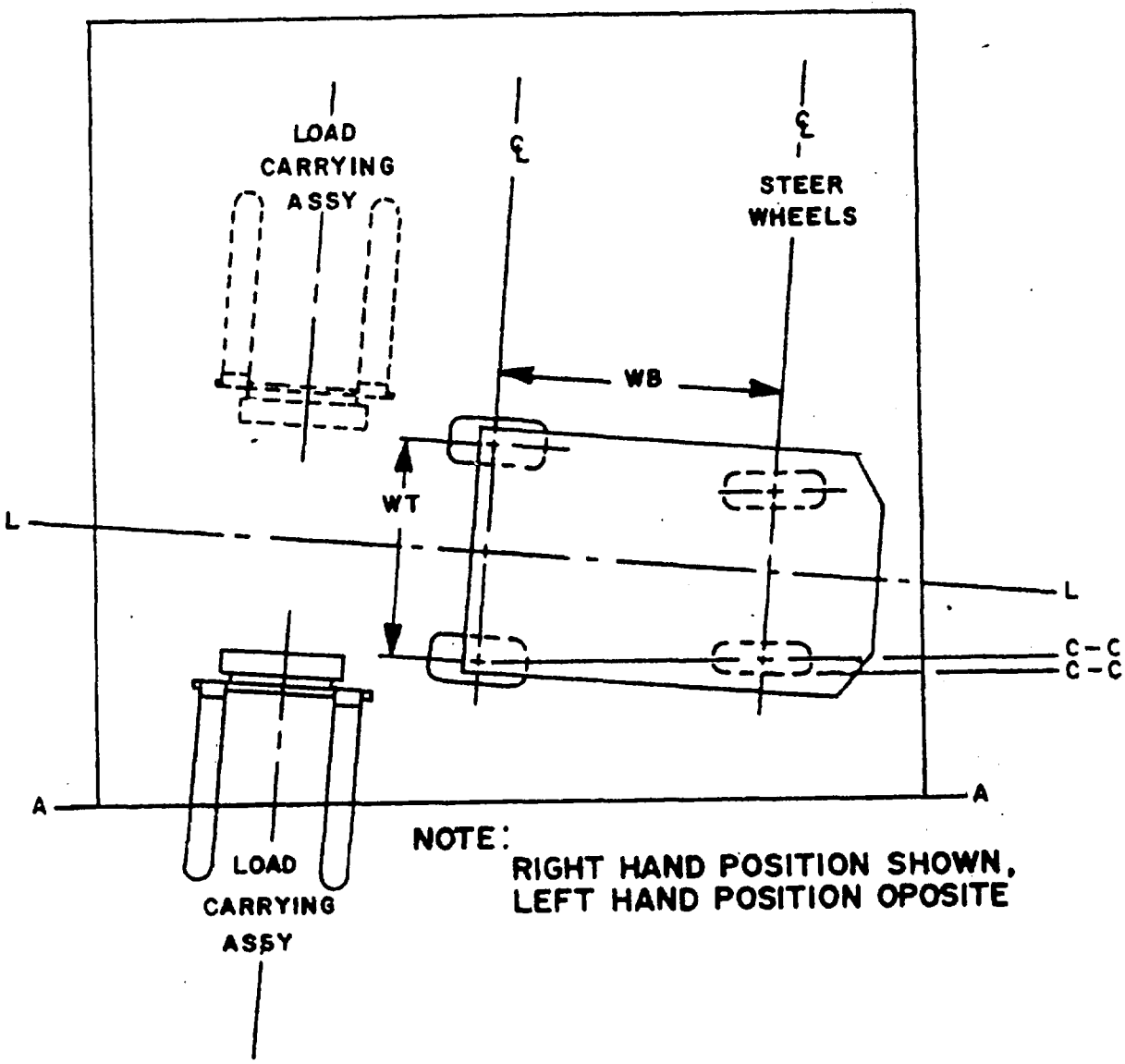
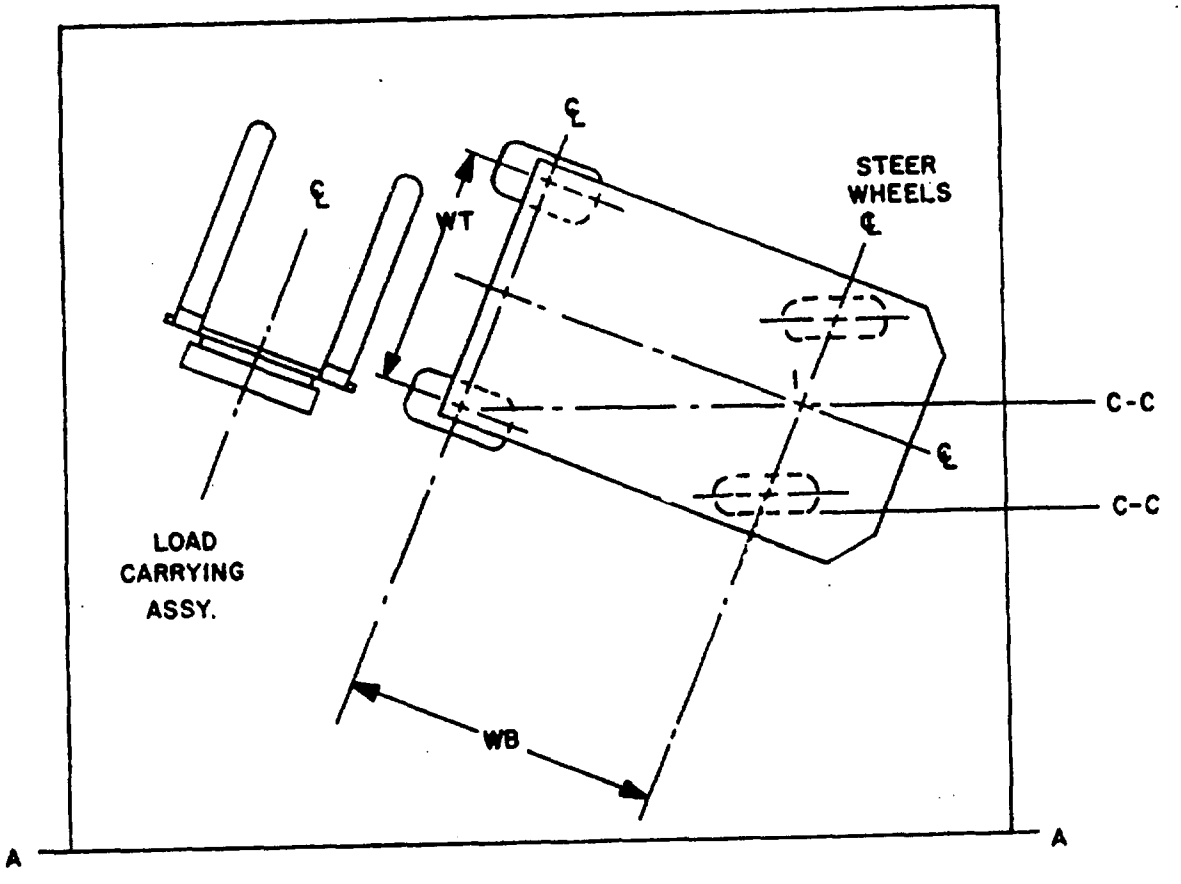


FIGURE A-9. Lateral stability-stacking. X-3991A
(non articulated steer axle)
(counterbalanced front/side loading operation).

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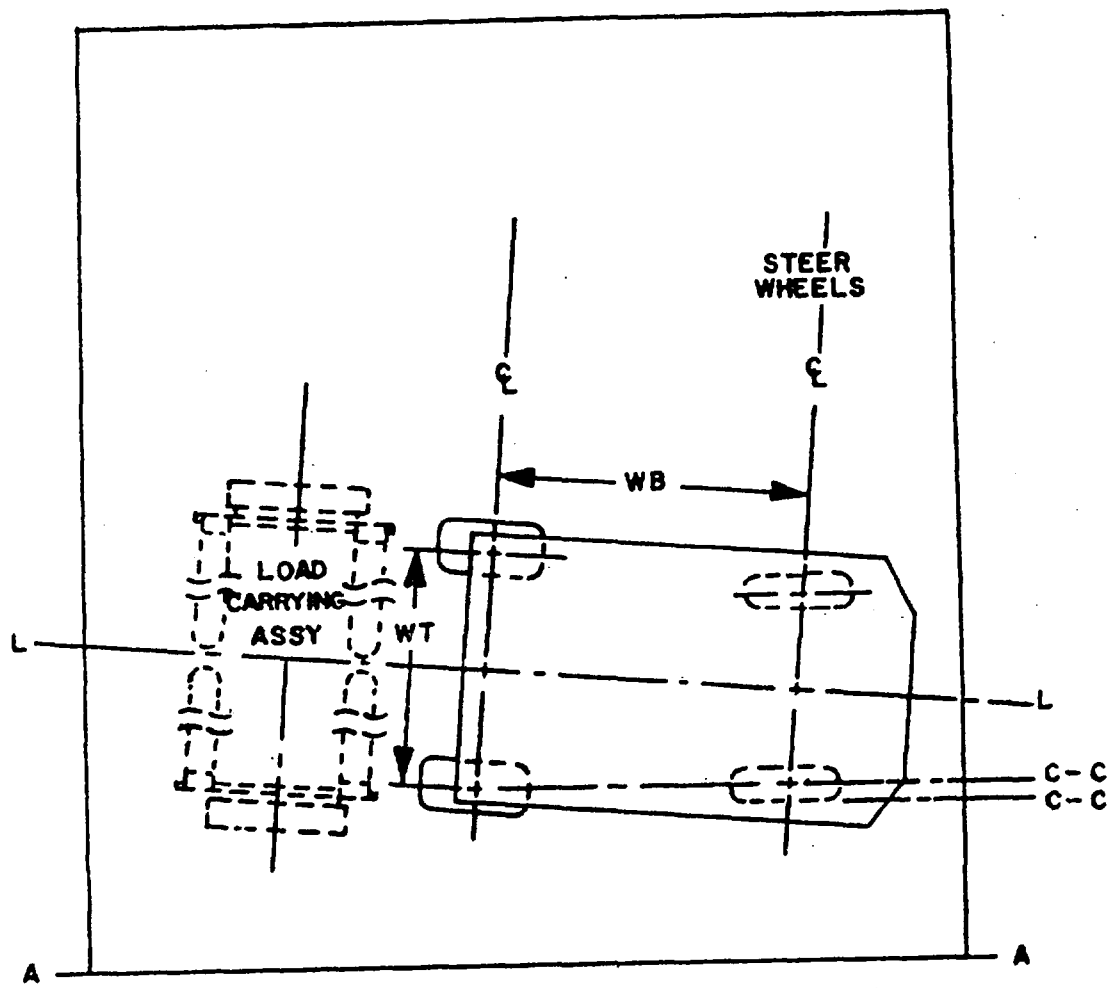
NOTE:

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X-3992A

FIGURE A-10. Lateral stability – traveling.
(articulated steer axle)
(counterbalanced front/side
loading operation.

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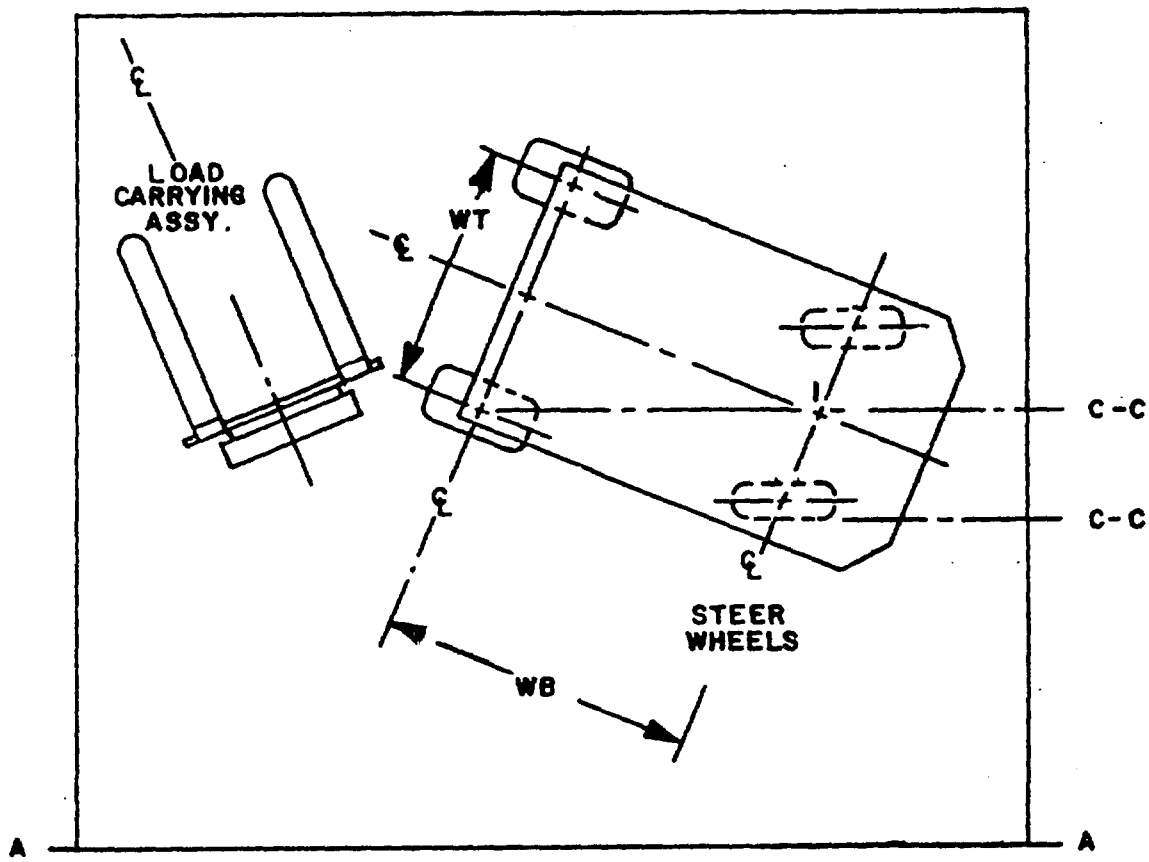


NOTE:
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FIGURE A-11. Lateral stability-traveling.
(non articulated steer axle)
(counterbalanced front/side loading operation).

X-3993A

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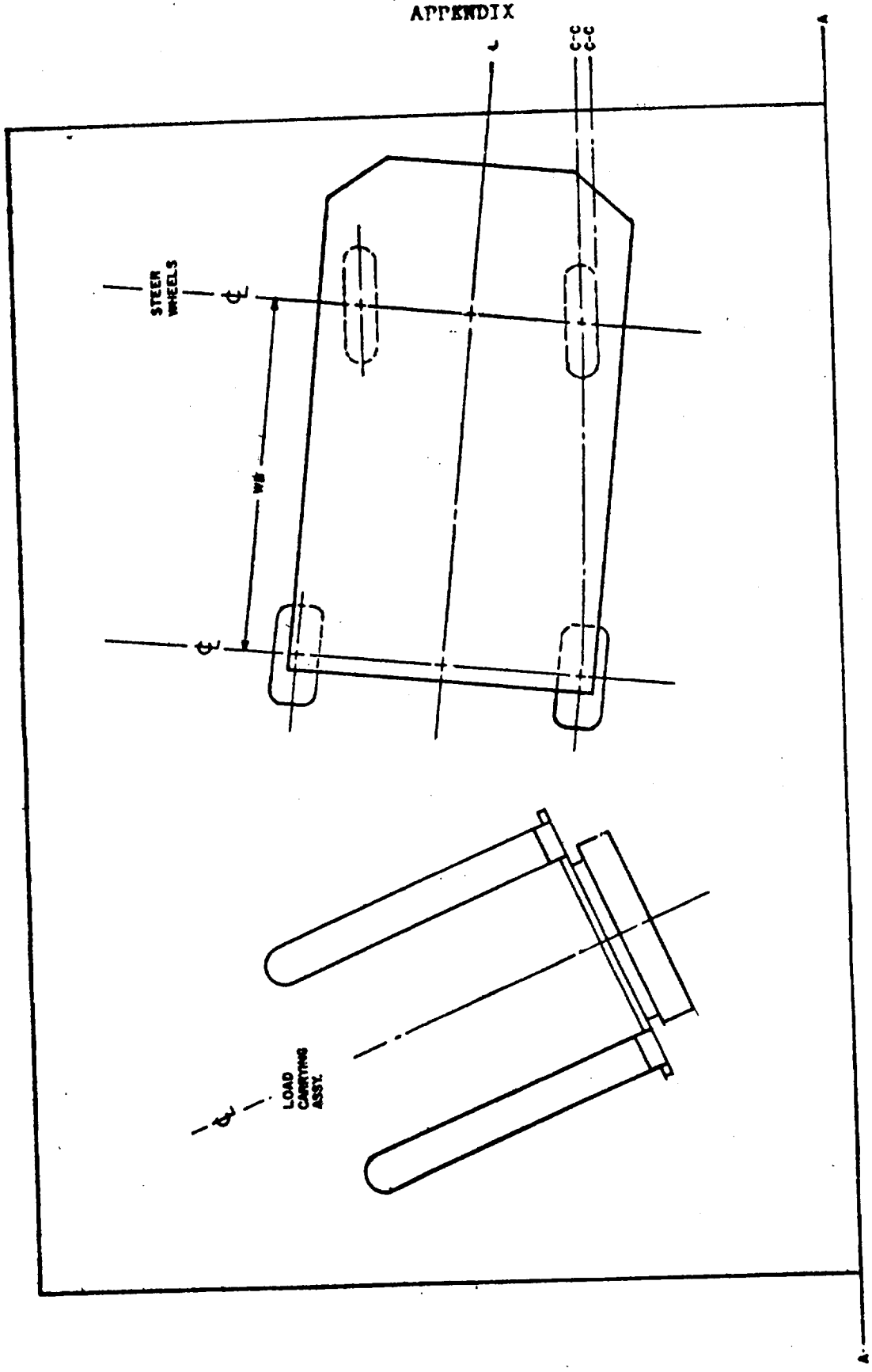


NOTE:

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FIGURE A-12. Lateral stability-stacking (L.H). X-3994A
(articulated steer axle)
(counterbalanced front / side loading operation).

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X3646

FIGURE A-13 Lateral stability - static.
(non-articulated steer axle)
(counterbalanced front/side loading operation)

NOTE:
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APPENDIX

TEST METHOD NO. 13

STOPPING DISTANCE

1. Test Course:

   (a) Test course shall be a dry, level surface of sufficient length to obtain maximum speed plus sufficient stopping distance.

2. Test Apparatus:

   (a) Tape measure.

3. Test Procedure:

   (a) With a 4000 pound load on the forks in the front carry position, run the truck in the rearward direction.
   (b) After reaching top speed, the brakes shall be applied.
   (c) Measure the distance traveled by the truck after application of the brake to the final stopping point. Record this distance.
   (d) Repeat (2) thru (c) for a total of three times. Stopping distance shall be the average of the three tests.

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APPENDIX

TEST METHOD NO. 14

TIRE LOADING

1. Test Apparatus:

(a) Platform scales.

2. Test Procedure:

(a) Front Carry Position

(1) Position truck forks in the front carry position along the longitudinal centerline of the truck.
(2) Determine the weight supported by each trail wheel without a load on the forks by driving the trail wheels onto a platform scale or jacking them up with two axle scales. When the trail wheels are weighed together, the total weight should be divided by the number of trail tires. The upright shall be in a vertical position and the forks approximately 12 inches above the floor.
(3) Determine the weight supported by the load wheels without a load on the forks by driving the load wheels on a platform scale or jacking them up with two axle scales. When the load wheels are weighed together, divide the total weight by the number of load-carrying tires. The upright shall be vertical, and the forks shall be approximately 12 inches above the floor.
(4) Repeat steps (2) and (3) with 4000 pound load on the forks centered on the carriage.

(b) Side Carry Position

(1) With a 4000 pound load on the forks pivot the forks to the side carry position and traverse the load such that the vertical face of the forks extends 2 inches beyond the right or left side of the truck. Determine the weight supported by the trail wheel and load wheel for the side of the truck which the load extends beyond. If pivot of 180 degrees is furnished determine the weight supported by both trail wheels and load wheels with the load pivoted in both directions. The upright shall be in the vertical position and the fork approximately 12 inches above the floor.
(2) With a 4000 pound load on the forks, pivot the forks 45 degrees and shift the load in the opposite direction. Determine the weight supported by the trail wheel and load wheel for the side of the truck that the load has been shifted to. The upright shall be in the vertical position and the forks approximutely 12 inches above the floor.

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APPENDIX

TEST METHOD NO. 15

POWER CONSUMPTION TEST

1. Test Course:

The test course shall consist of three load stations located approximately 30 feet apart and a level return route of approximately one hundred feet for a total course length of approximately 160 feet. Load station 1 shall be at floor level. Load station 2 shall be a load platform approximately 48 inches above floor level. Station 3 shall be a load platform approximately 90 inches above floor level. Two (2) 4000 pound test loads shall be placed; one at station 1 on the floor and one at station 3 on the load platform.

2. Test Apparatus:

Stop Watch

3. Test Procedure:

(a) Starting at a point just before station 1 proceed to station 1 and pick up the load. Back the truck to station 2, raise the load and place it on on the platform. Proceed forward to station 3 and pick up the load there. Travel forward with this load to station 1 and place the load on the floor. Back the truck to station 2 and pick up the load previously placed there. Proceed forward to station 3 and place the load on the platform. Travel without load to starting point. Stop the truck; get up from the seat; sit sit down; sound the horn; turn lights on, then off; turn key switch off, then on. This completes one cycle of the course.

(b) The test course shall be negotiated continuously for 6-1/2 hours and at a rate of approximately 15 cycles per hour. One battery only shall be used. This battery shall not be recharged during the test. Other than momentary stops above, truck shall only be stopped to change operators. Time for changing operators shall not be counted as part of test time. Total test time shall be recorded.

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APPENDIX

TEST METHOD NO. 16

RAIN

1. Test Apparatus:

   (a) Timing device.
   (b) Rain gage (U. S. Weather Bureau type preferred).
   (c) Spray fixture.

2. Test Procedure:

   (a) Subject the truck to a simulated rainfall of not less than 4 inches per hour. The simulated rainfall shall impinge on the truck at an angle of 45 degrees, plus or minus 5 degrees, and impinge with a force equivalent to a 30-mile per hour wind. The simulated rainfall shall disperse uniformly over the surfaces of the truck.
   (b) Expose the top and left side of the truck to the rain described above for a period of 5 minutes.
   (c) Repeat (b) with the top and front exposed.
   (d) Repeat (b) with top and right exposed.
   (e) Repeat (b) with top and rear exposed.
   (f) Examine all instruments and electric controls for evidence of moisture.
   (g) Demonstrate operation of the truck.

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APPENDIX

TEST METHOD NO. 17

RELIABILITY TEST

1. Test Course:

(a) Layout of course. The test course shall be set up equivalent to the provisions of Figure A-14. The course perimeter shall be demarcated with suitable barriers or indicators. Rubber pylons or other suitable marking shall be placed on all corners and other critical points to assist in keeping the truck being tested within the course limits.

(b) Obstacle inclusion. The obstacle shall be in accordance with the dimensions shown on Figure A-15 and positioned on the test course as shown on Figure A-15. Iron, wood, cement or steel blocks may be used.

(c) Ramp inclusion. The course shall include a ramp constructed in accordance with the provisions of Figure A-16.

(d) Course surface. The course shall be paved with concrete having a fine float finish, or equivalent, having a road resistance of approximately 30-50 pounds per ton. It shall be dry, clean and free of any nonplanned obstacles or foreign material while conducting the reliability test.

(e) Pallet areas. Pallet loads shall be stacked in accordance with Figure A-17, A-18, and A-19 and shall be a 48 inch cube.

2. Test Procedure:

The test shall consist of 2000 cycles in Rack Storage Operation (3000 pound capacity test load) and 2000 cycles in Bulk Storage Operation (4000 pound capacity test load). The rack storage operation and bulk storage operation tests shall be conducted as follows:

(a) The light switch shall be turned "OFF" and "ON" at the beginning of each lap and upon entering each pallet area.

(b) The truck under test shall begin each lap at the point labeled "START" on Figure A-14. As the operator starts the truck, he shall operate the horn for approximately 1 second. On lap number 1, the truck shall proceed in a forward direction, following the centerline along the portion of the course marked "A" until it reaches the first pallet area. The load will have been placed in this location on the stack marked "X".

(c) When retrieving and placing loads in position 1, 2, and 3, all motions of the truck and loading devices may be employed.

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APPENDIX

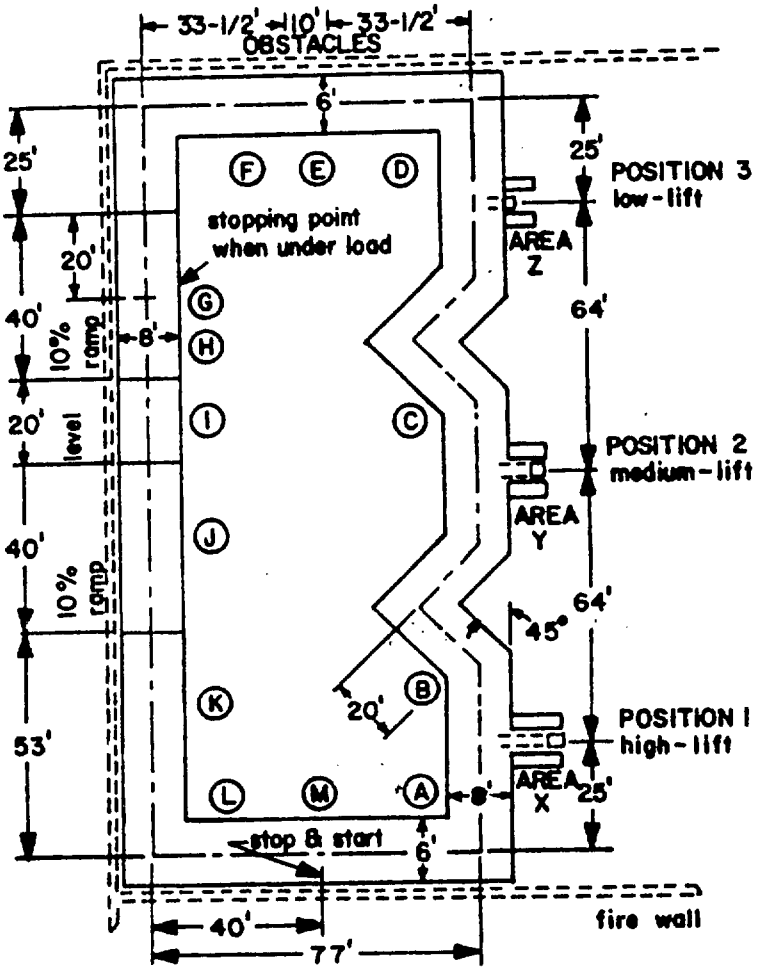
TEST METHOD NO. 17 (CONT'D)

(d) The truck shall retrieve the load from position 1 area "X" proceed to position 2 area "Y" and place the load in the prescribed position, then proceed unloaded to position 3 area "Z" and retrieve the load located at this position. The truck shall continue around the course, negotiating the obstacles, proceeding to the mid point on the ramp slope. At this point the truck shall be stopped. The parking brake shall be activated and shall hold the truck for one second. The truck shall resume travel up the ramp and continue negotiation of the course back to the start point. The master switch shall be turned off. This constitutes one cycle. The course shall then be negotiated in the same manner with the exception that the load handling sequence shall be opposite that of the preceding cycle. Continue for 2000 cycles.

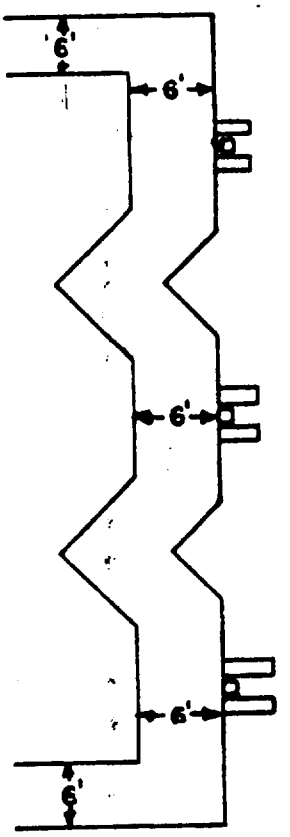
(e) The "Hourly Time Record Sheet" shall be filled out for each operating hour of the performance test. Average lap speed per hour shall be recorded.

(f) Maintenance inspections shall be performed in accordance with the maintenance service and inspection sheet.

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(BULK STORAGE)



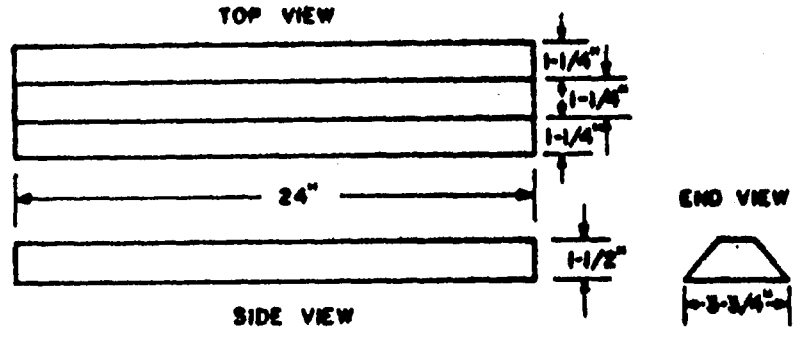
(RACK STORAGE)

LEGEND:

Ⓐ, ECT. = Section designation

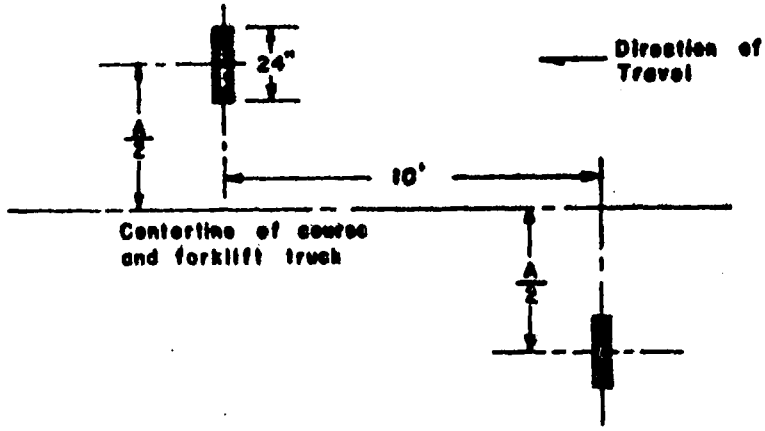
FIGURE A-14. Typical indoor test course. X-2309D

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2 Blocks required
Blocks to be attached to test course as shown below

LAYOUT OF OBSTACLE COURSE



A = Distance between centerlines of driving wheels

FIGURE A-15. Details of obstacle construction for truck, fork, lift, solid rubber tires.

X-2310A

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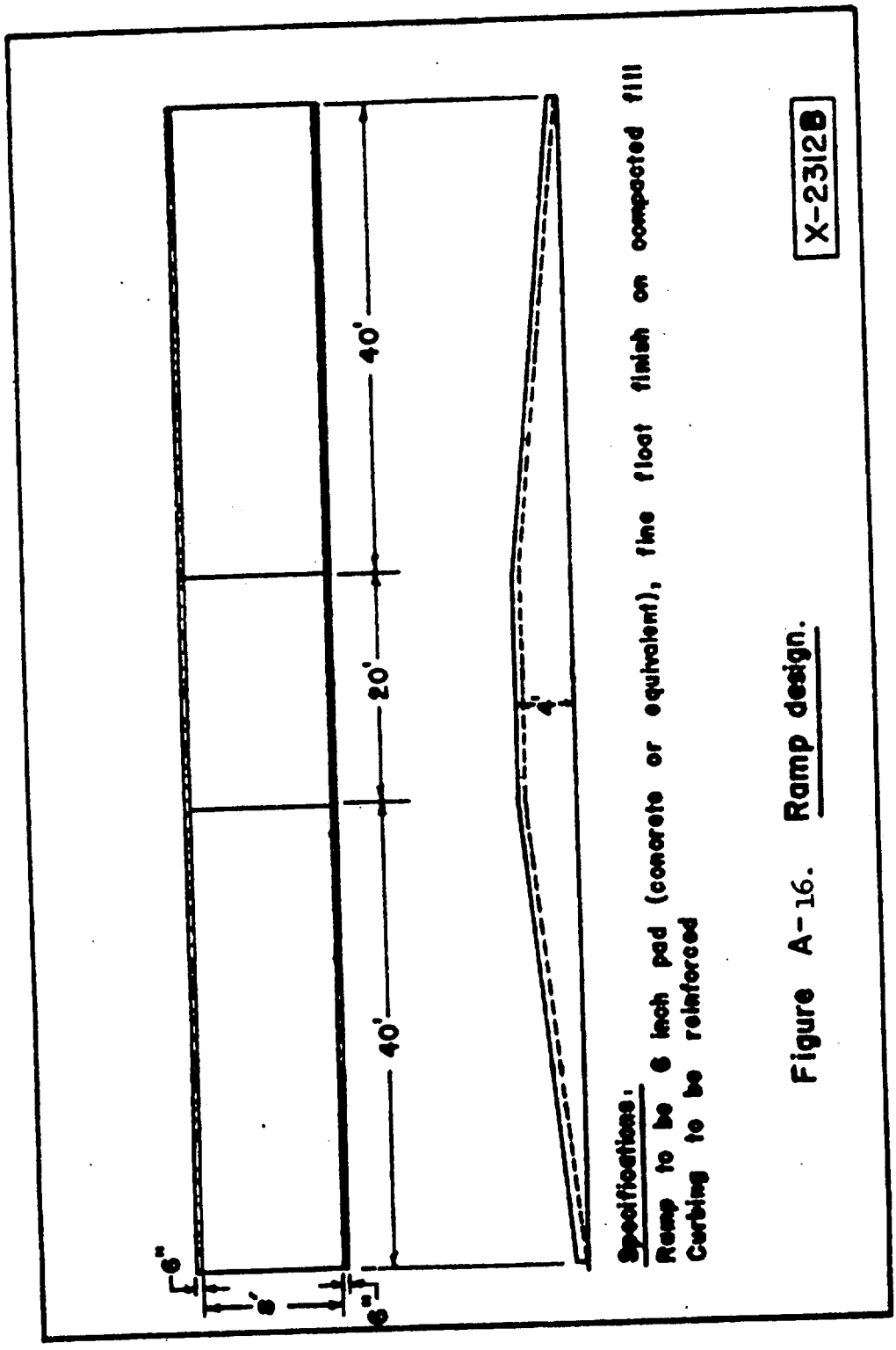
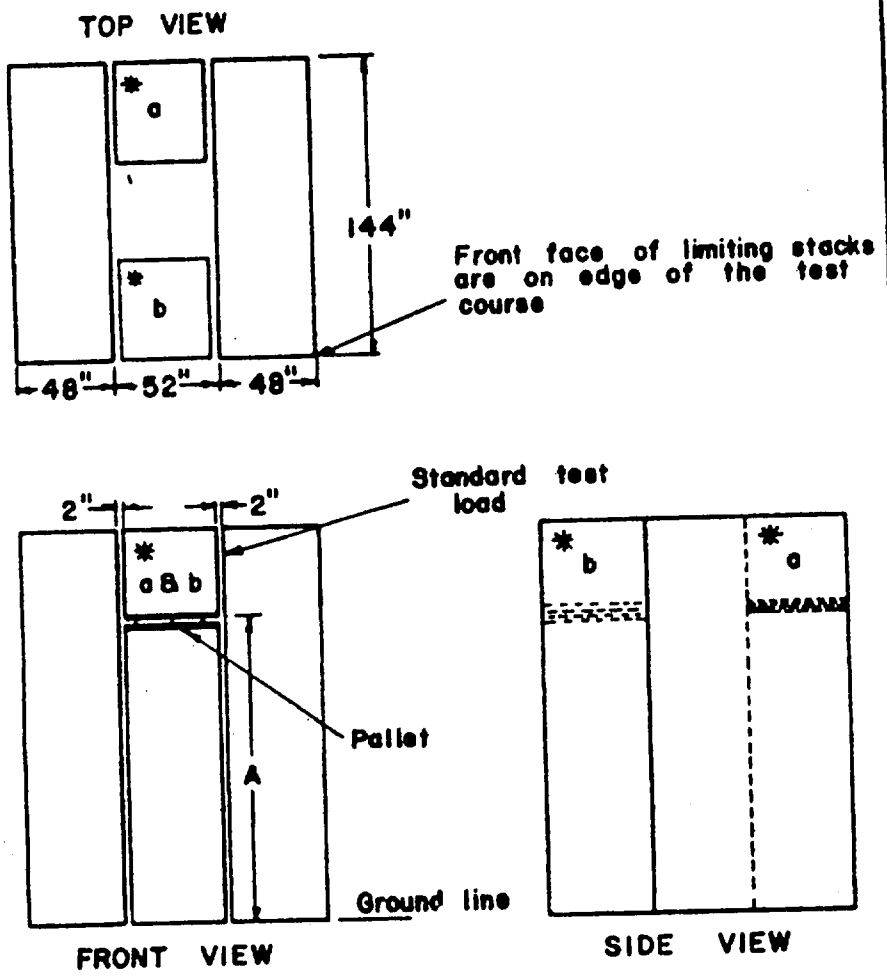


Figure A-16. Ramp design.

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A = Maximum lift height for forks less 6 inches

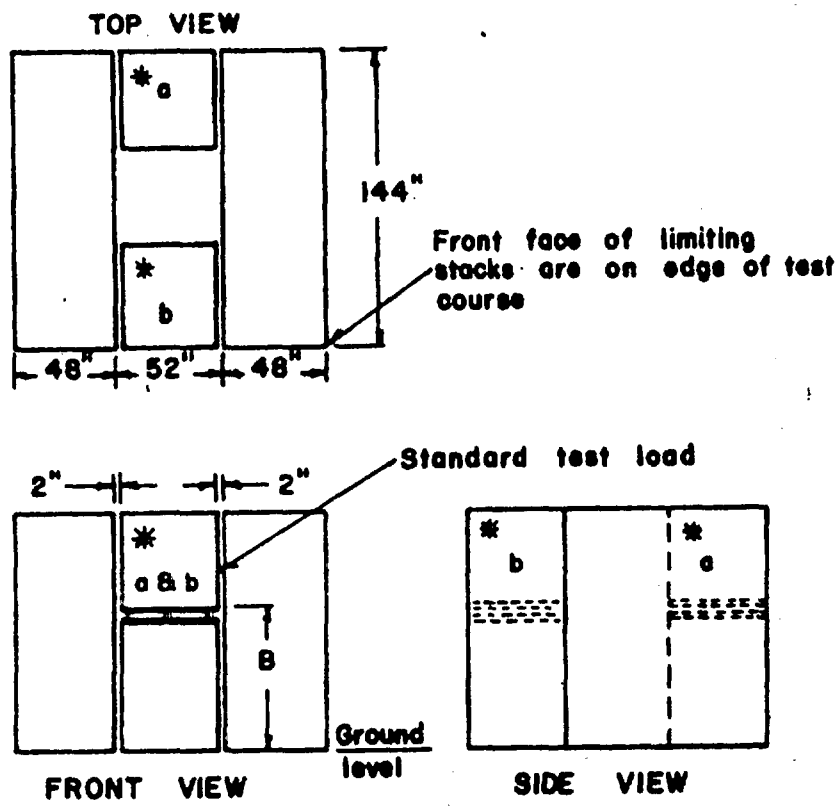
Support for standard load should be of suitable strength to sustain test load. Limiting stacks may be pallets, pallet racks or both.

* (a) BULK STORAGE
(b) RACK STORAGE

FIGURE A-17. Stack layout-position 1.

X-2313B

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B = One-half maximum lift height of forks

Support for standard load should be of suitable strength to sustain test load. Limiting stacks may be pallets, pallet racks, or both.

* (a) BULK STORAGE
(b) RACK STORAGE

FIGURE A-18. Stack layout-position 2

X-2315A

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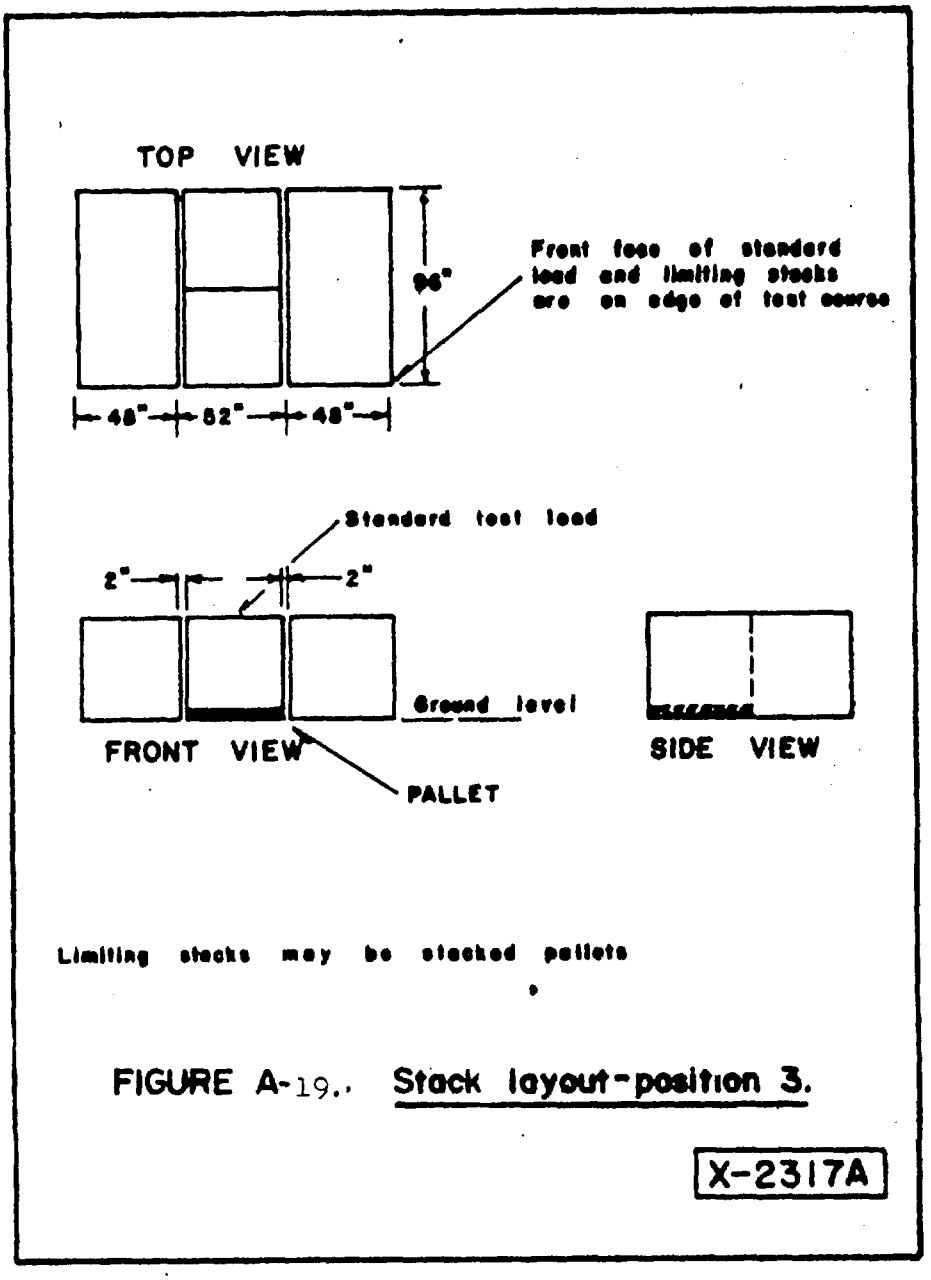


FIGURE A-19.. Stack layout-position 3.

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APPENDIX

MAINTENANCE SERVICE AND INSPECTION SHEET

MATERIALS HANDLING EQUIPMENT-ELECTRIC

The following inspections and checks shall be made prior to endurance testing and prior to the post testing. Items 1 through 12 shall be checked after completion of each 2000 circuits of the endurance test.

1.	Check all instruments for proper operation - check free play in steering.
2.	Check brake pedal travel - check fluid level - adjust brakes if necessary - check safety seat brake - check linkage and operation - remove all foreign matter from tires - brake lining and drums, check wheel cylinders.
3.	Inspect all lights.
4.	Check battery - water level and specific gravity.
5.	Check hydraulic oil dipstick - add oil if necessary.
6.	Inspect all safety equipment - overhead guards; check for cracked or bent forks.
7.	Check controller contact tips, switches.
8.	Check motors' brush wear, spring tension, commutator wear - blow out carbon dust.
9.	Check cylinders for leaks and wear - wipe off.
10.	Inspect all wires and terminals; tighten and replace where necessary.
11.	Lubricate chassis using manufacturer's reference charts - replace missing or broken fittings where necessary - make all necessary adjustments.
12.	Inspect drive unit - check for noise - check oil level - fill if necessary - do not overfill.
13.	Inspect steering gear - check oil level - fill if necessary - do not overfill.
14.	Inspect carriage and channel assembly (rollers, carriage and mast) for wear and damage - adjust and aline if necessary.

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APPENDIX

MAINTENANCE SERVICE AND INSPECTION SHEET (CONT'D).

15.	Inspect lift chains for wear and damage.
16.	Wheel bearings - check and adjust, tighten all hub flange bolts.
17.	Hydraulic system - check for noise, leaks, and operation - tighten all lines and fittings.

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APPENDIX

HOURLY TIME RECORD SHEET

Date_____

Truck Manufacturer_____ Clock Readings:_____

Model_____ Test Location_____ Start_____

Serial No._____ Recorded by_____ Finish_____

Time	Laps	Average Lap Time	Comments

Instructions:

1. Start decimal minute stop watch at zero at beginning of each hour.
2. Record watch reading at end of each hour and record elapsed time.

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APPENDIX

TEST METHOD NO. 18

LOAD BACK REST

1. Test Course:

   (a) Shop area.

2. Test Apparatus:

   (a) Test weight.
   (b) Stop watch.

3. Test Conditions: A completely assembled truck may be used for this test or it may be performed as a bench test using fork carriage and back rest. A force equal to the rated load shall be applied at right angles to the back rest and uniformly distributed over the entire area comprising the back rest.

4. Test Procedure:

   (a) Place a metal sheet over the total area comprising the back rest.
   (b) Apply a weight equal to the rated load, at right angle to the back rest area.
   (c) Hold weight on truck's load back rest for not less than 3 minutes.
   (d) Release pressure and inspect the entire back rest, for cracks, breaks, failures or any permanent deformation of any members of the back rest.
   (e) Check back rest for ease of removal after loading.

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APPENDIX

TEST METHOD NO. 19

HYDRAULIC SYSTEM CLEANLINESS

1. Test Apparatus:

   (a) Liquid Automatic Particle Counter.
   (b) A microscope and appropriate grid, as called out in ASTM-F-24-65, to get the same readings as using a particle counter calibrated using "AC" fine test dust (ANSI B93.28).

2. Test Procedure:

   (a) When hydraulic line volume exceeds 1/2 the volume of the related cylinder, connect the hydraulic lines on the cylinders to bypass related components.
   (b) Run the pump pumping the oil through each circuit for not less than 5 minutes. Perform the particle count as described in (d) below.
   (c) Connect each line to its related component. Operate each previously bypassed function of the system for 5 minutes continuously through full stroke and at maximum rates. Perform the particle count as described in (d) below.
   (d) The procedure for determining contamination shall be as follows:

      (1) Using Test Apparatus Under (a)
      Contamination levels shall be determined by particle counts. The particle counter used for evaluation of the samples shall be calibrated in accordance with ANSI Recommended Standard Method for Calibration of Liquid Automatic Particle Counters using "AC" Fine Test Dust (ANSI B93.28). The required counts shall be the average of not less than three consecutive counts. Samples may be taken from the reservoir or upstream of the filter(s) used for cleanup.

      (2) Using Test Apparatus Under (b)
      Take 3 samples of oil from each truck and average count. Samples may be taken from the reservoir or upstream of the filter(s) used for cleanup.

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APPENDIX

TEST METHOD NO. 20

CYLINDER PROOF

1. Test Apparatus:

(a) Hydraulic test bench or the truck capable of developing two times system working pressure.
(b) Pressure gage.
(c) Thermocouple.

2. Test Procedure:

(a) Double acting cylinder.

(1) Bottom piston in cylinder barrel.
(2) Fill both sides of the piston with oil at a minimum of 120° F.
(3) With the piston end port capped, apply an oil pressure equal to 1.5 times the working pressure to the rod end of the piston for a minimum of 60 seconds. Release pressure.

(b) Single acting cylinder.

(1) Extend cylinder to maximum extension by filling with oil at a minimum of 120° F.
(2) Apply oil pressure equal to 1.5 times the working pressure at the inlet port for a minimum of 60 seconds. Release pressure.

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APPENDIX

TEST METHOD NO. 21

HYDRAULIC FITTINGS

1. Test Apparatus:

   (a) Common handtools.

2. Test Procedures:

   (a) Remove one hydraulic hose assembly which has permanent type fittings.
   (b) Replace the hose assembly with an equivalent field attachable reuseable fitting only to assure interchangeability.
   (c) Repeat (a) and (b) above until each hose assembly with permanent type fittings is replaced with a hose assembly with reuseable fittings.
   (d) After every hose assembly with permanent fittings has been replaced with a hose assembly with reuseable fittings, reinstall the original hose assemblies on the truck.

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APPENDIX

TEST METHOD NO. 22

UPRIGHTS

1. Test Apparatus:

The test apparatus shall consist of a hydraulic test bench capable of providing the necessary hydraulic power and control to perform the required testings. The test upright may be mounted on a fixture or on a truck.

2. Test Conditions:

(a) Oil temperature shall not exceed 225° F.
(b) Upright lubrication shall be performed at not less than 1000 test cycles with lubricants specified in the applicable truck technical manuals.
(c) All tests shall be conducted at a lifting speed and lowering speed equal to the lifting and lowering speeds specified for the truck. Speeds may vary plus or minus 5 feet per minute.

3. Test Procedures:

The upright shall be subjected to 35,000 cycles as specified below:

(a) Cycle consists of lifting to not less than 3 inches of maximum fork height and lowering to not more than 3 inches above ground level of the upright in the vertical to 2 degrees backtilt position with the loads specified:

(1) 20,000 cycles with rated load centered.
(2) 5000 cycles with rated load 4 inches off center to the right.
(3) 5000 cycles with rated load 4 inches off center to the left.
(4) 5000 cycles with rated load plus 10 percent centered.

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